Chapter 4

Molecular Reaction Dynamics

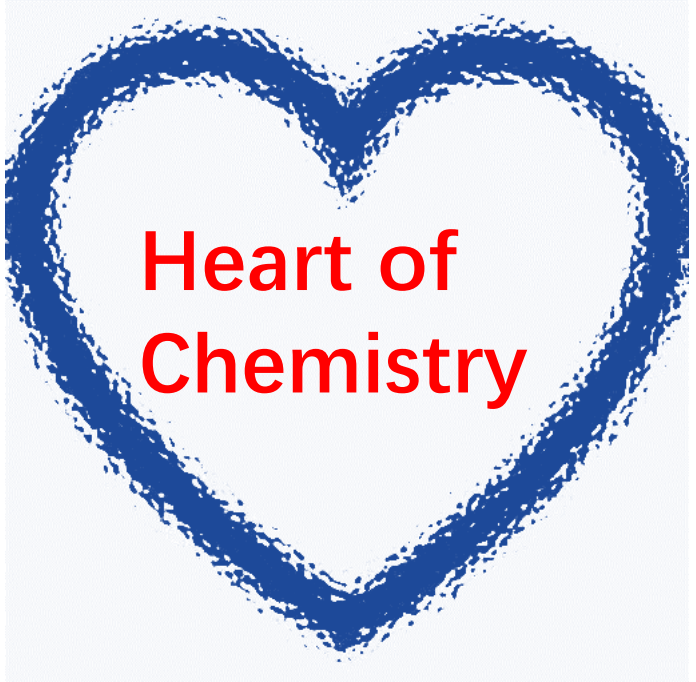
(分子反应动态学)

Contents

- Collision Theory
- Transition State Theory
- Electron Transfer

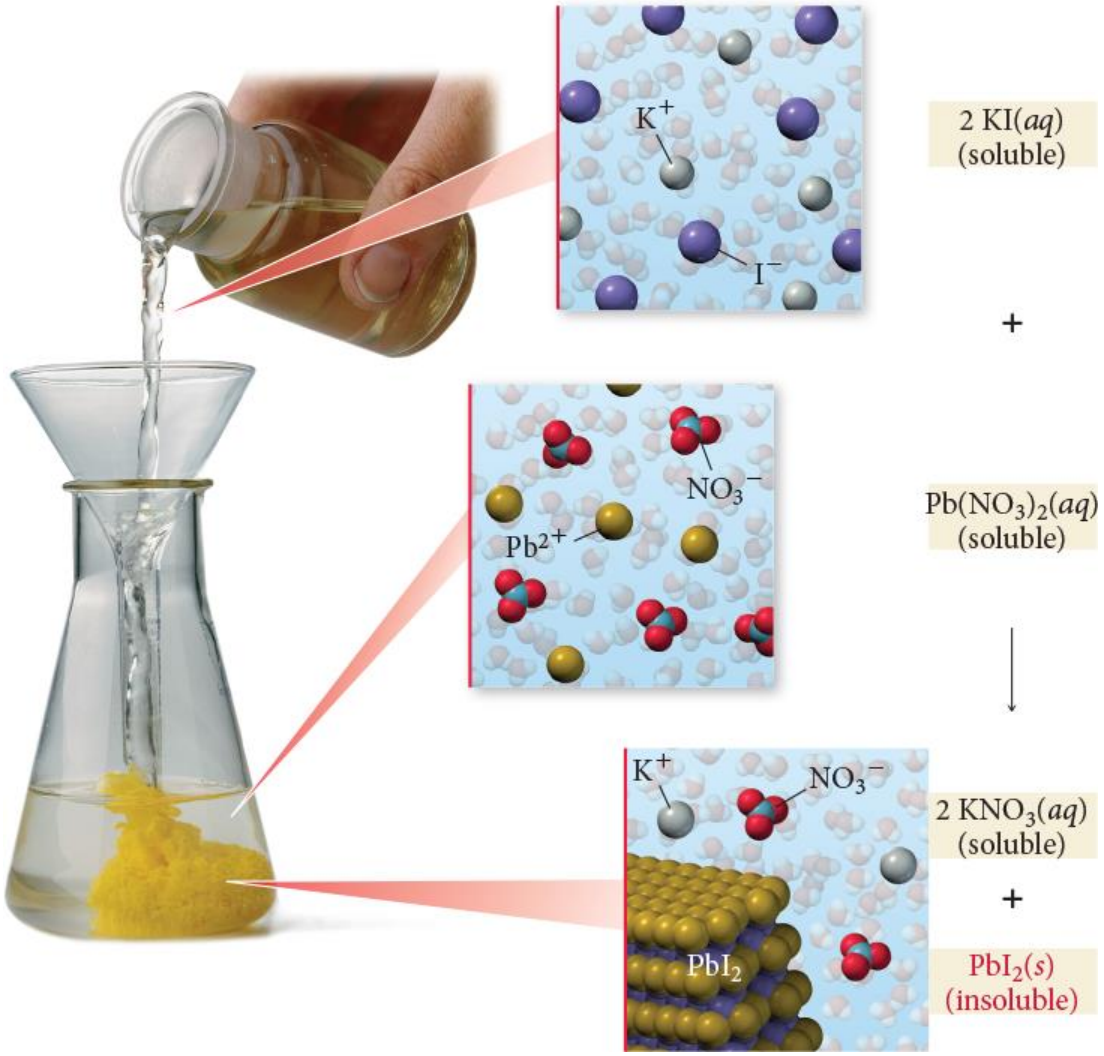
Contents

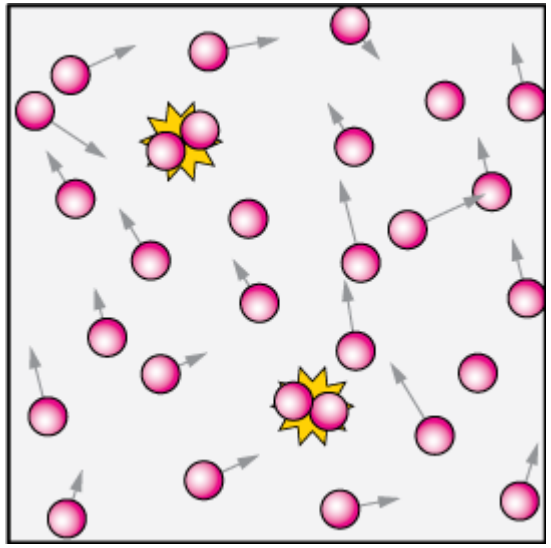
- Collision Theory
- Transition State Theory
- Electron Transfer



☼ What happens to molecules at the climax of reactions?
☼ Why reaction rates differ for different reactions?
☼ How to **estimate rate constant** of an elementary reaction?

Reactive encounters



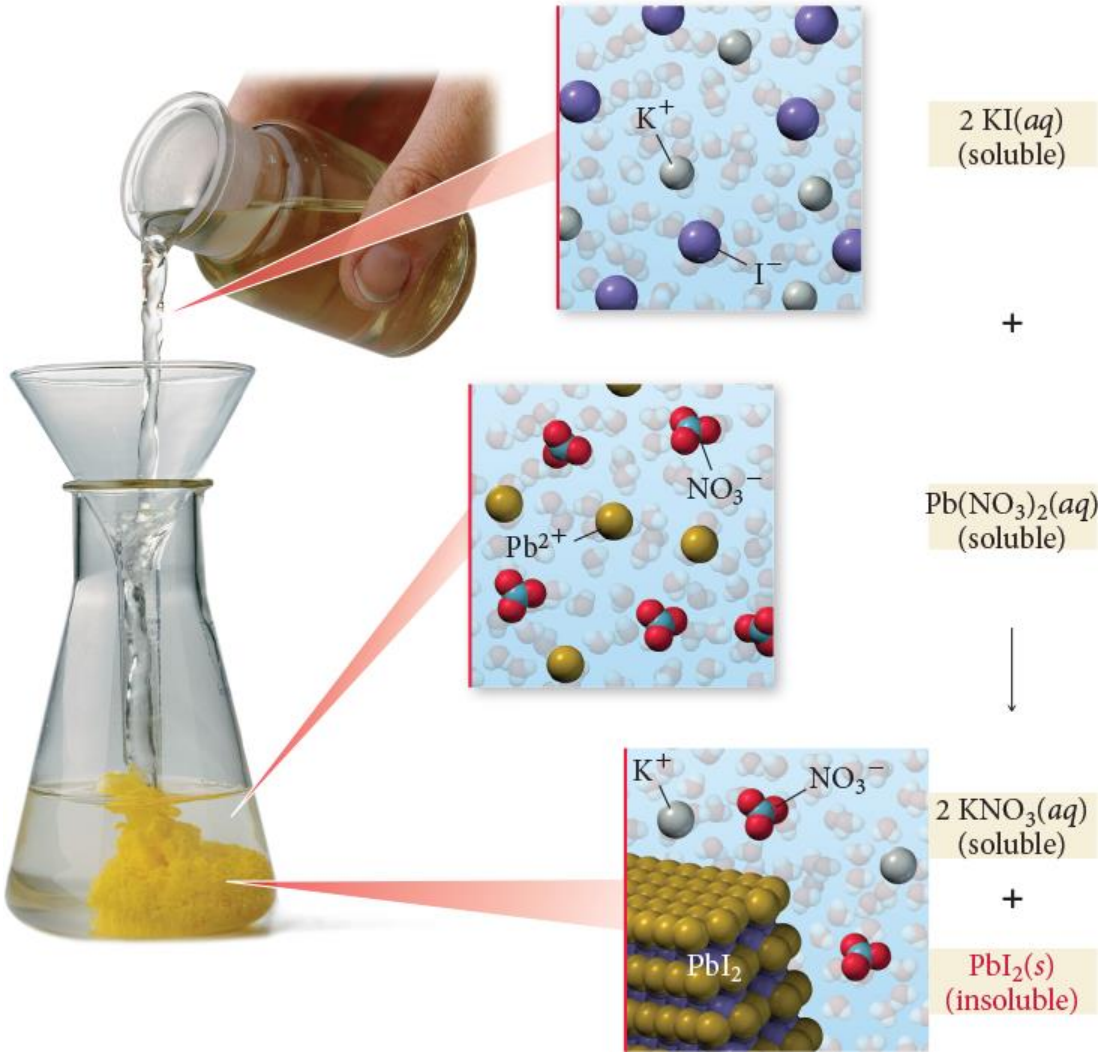


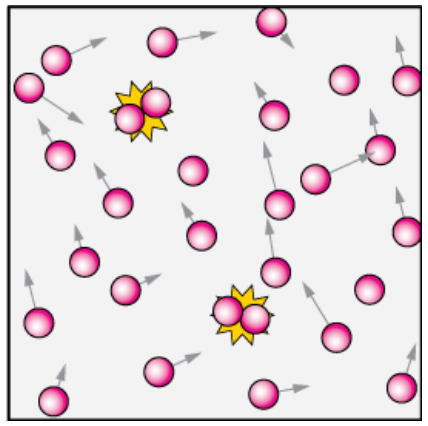
In gas-phase

In solution →

Reactants must meet and have a certain minimum energy

Reactive encounters





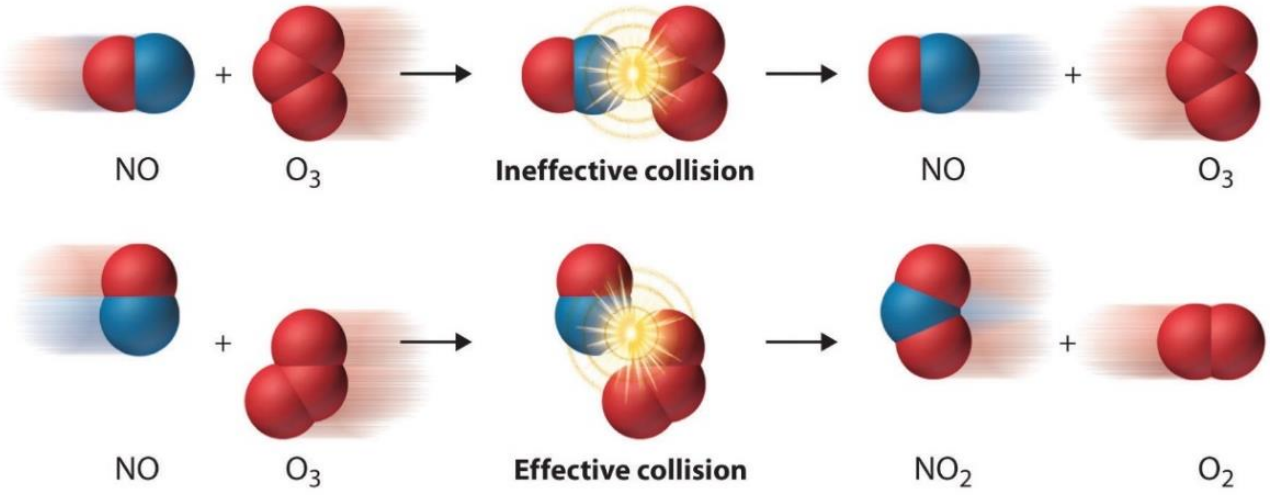
In gas-phase

Effective collision:

Collide

Enough energy

Correct orientation



In solution

Diffusion-controlled

Diffuse together, then get energy from surroundings

Collision Theory

$$A + B \rightarrow P \qquad v = k_2[A][B]$$

- **Rate of collisions**
  - Mean speed of molecules, $\bar{c}$
  - Collision cross-section, σ
  - Number densities of A and B

$$v \propto \sigma \overbrace{(T/M)^{1/2}}^{\propto \bar{c}} \mathcal{N}_A \mathcal{N}_B \propto \sigma (T/M)^{1/2}[A][B]$$

- **Activation energy E_a**

$$k_2 \propto \sigma (T/M)^{1/2}$$

$$k_2 \propto \sigma (T/M)^{1/2} e^{-E_a/RT}$$

- In a certain relative **orientation**

$$k_2 \propto P\sigma (T/M)^{1/2} e^{-E_a/RT}$$

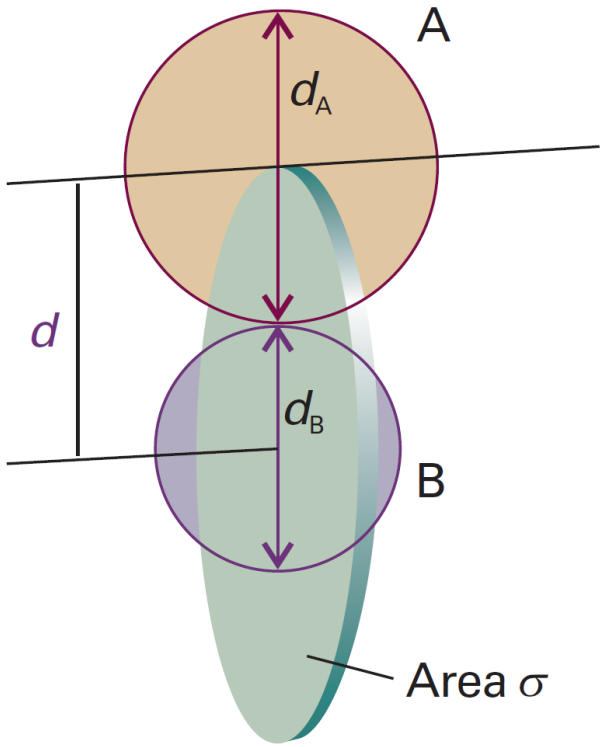
Steric factor

Collision Theory

$$A + B \rightarrow P \qquad v = k_2[A][B]$$

For a successful collision, $\boldsymbol{k_2} \propto$

- Encounter rate
- Minimum energy requirement
- Steric requirement



$$\sigma = \pi d^2$$

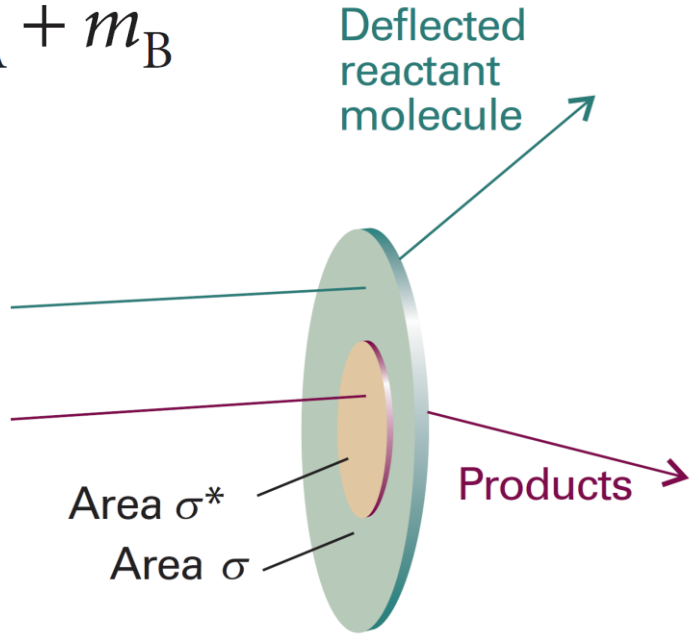
$$d = \tfrac{1}{2}(d_A + d_B)$$

Boltzmann distribution

$$k_2 = P\sigma\left(\frac{8kT}{\pi\mu}\right)^{1/2} N_A e^{-E_a/RT}$$

Avogadro's constant

$$\mu = \frac{m_A m_B}{m_A + m_B}$$



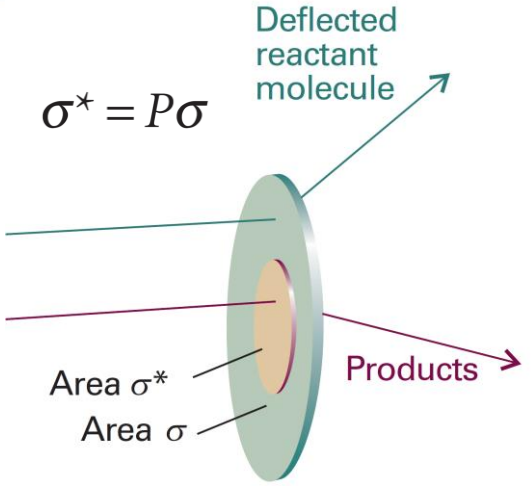
Reactive cross-section

$$\sigma^* = P\sigma$$

Steric factor P???

E_a???

$$k_2 = P\sigma\left(\frac{8kT}{\pi\mu}\right)^{1/2} N_A e^{-E_a/RT}$$



The Arrhenius form $k_2 = Ae^{-E_a/RT}$

Pre-exponential factor

***P* is normally <<1**

	$A/(\mathrm{dm^3\,mol^{-1}\,s^{-1}})$			
	Experiment	Theory	$E_a/(\mathrm{kJ\,mol^{-1}})$	*P*
$2\,NOCl \rightarrow 2\,NO + 2\,Cl$	9.4×10^9	5.9×10^{10}	102	0.16
$2\,ClO \rightarrow Cl_2 + O_2$	6.3×10^7	2.5×10^{10}	0	2.5×10^{-3}
$H_2 + C_2H_4 \rightarrow C_2H_6$	1.24×10^6	7.4×10^{11}	180	1.7×10^{-6}
$K + Br_2 \rightarrow KBr + Br$	1.0×10^{12}	2.1×10^{11}	0	4.8

Example: Estimating a steric factor

Estimate the steric factor for the reaction $H_2 + C_2H_4 \rightarrow C_2H_6$ at 628 K given that the pre-exponential factor is $1.24 \times 10^6\ dm^3\ mol^{-1}\ s^{-1}$.

$$\sigma(H_2) = 0.27\ nm^2$$

$$\sigma(C_2H_4) = 0.64\ nm^2$$

$m_1 = 2.016$ u for H_2 and $m_2 = 28.05$ u for C_2H_4

1 u $= 1.66054 \times 10^{-27}$ kg

Example: Estimating a steric factor

Estimate the steric factor for the reaction $H_2 + C_2H_4 \rightarrow C_2H_6$ at 628 K given that the pre-exponential factor is 1.24×10^6 dm^3 mol^{-1} s^{-1}.

$$A = \sigma \left(\frac{8kT}{\pi\mu} \right)^{1/2} N_A$$

$\sigma(H_2) = 0.27$ nm^2

$\sigma(C_2H_4) = 0.64$ nm^2

Take average → $\sigma = 0.46$ nm^2

$m_1 = 2.016$ u for H_2 and $m_2 = 28.05$ u for C_2H_4

1 u = 1.66054×10^{-27} kg

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = 3.12 \times 10^{-27} \text{ kg}$$

$$\left(\frac{8kT}{\pi\mu} \right)^{1/2} = 2.66 \times 10^3 \text{ m s}^{-1}$$

Theoretically

$$A = \sigma \left(\frac{8kT}{\pi\mu} \right)^{1/2} N_A = 7.37 \times 10^{11} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

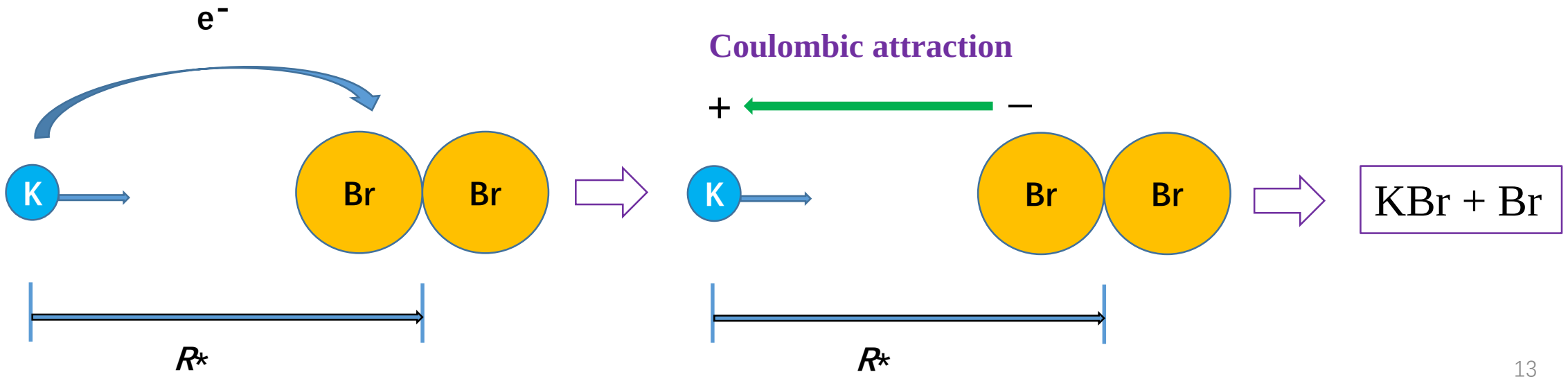
→ $P = 1.7 \times 10^{-6}$

Harpoon Reaction

	$A/(\mathrm{dm^3\ mol^{-1}\ s^{-1}})$			
	Experiment	Theory	$E_a/(\mathrm{kJ\ mol^{-1}})$	P
$K + Br_2 \rightarrow KBr + Br$	1.0×10^{12}	2.1×10^{11}	0	4.8

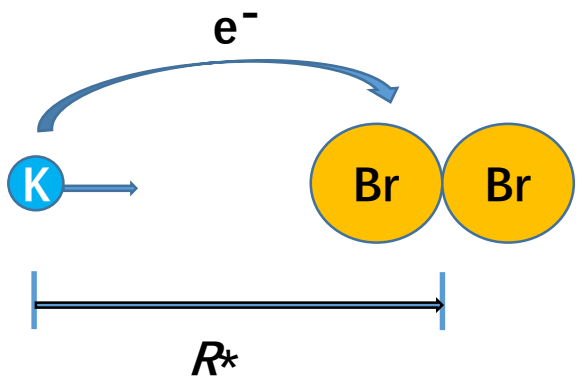


Harpoon mechanism: Electron transfer preceded the atom extraction. **It extends σ.**



Harpoon Reaction

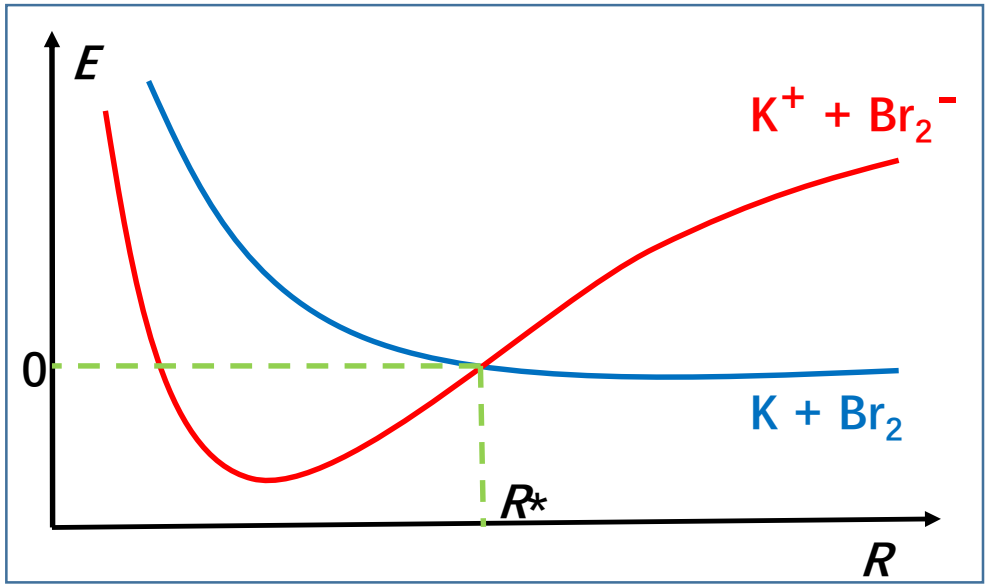
Harpoon mechanism: Electron transfer preceded the atom extraction. **It extends σ**.



$$P = \frac{\sigma^*}{\sigma} = \frac{R^{*2}}{d^2}$$

R*???





Coulombic interaction energy

$$E = I - E_{ea} - \frac{e^2}{4\pi\varepsilon_0 R}$$

= 0 (R=R*)

I: Ionization energy of K

E_{ea}: Electron affinity of Br_2

Example: Estimating a steric factor 2

Estimate the value of P for the harpoon mechanism by calculating the distance at which it becomes energetically favourable for the electron to leap from K to Br_2.

$$E = I - E_{ea} - \frac{e^2}{4\pi\varepsilon_0 R}$$

At $R = R^*$, $E = 0$ ⟶

$$\frac{e^2}{4\pi\varepsilon_0 R^\star} = I - E_{ea}$$

$$P = \frac{\sigma^\star}{\sigma} = \frac{R^{\star 2}}{d^2} = \left\{ \frac{e^2}{4\pi\varepsilon_0 d(I - E_{ea})} \right\}^2$$

$$d = R(\mathrm{K}) + R(\mathrm{Br_2}) = 400\ \mathrm{pm}$$

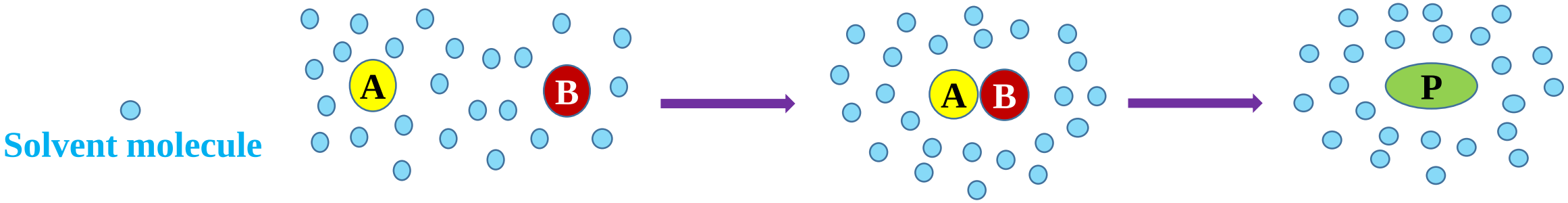
$$I = 420\ \mathrm{kJ\ mol^{-1}}$$

$$E_{ea} \approx 250\ \mathrm{kJ\ mol^{-1}}$$

⟹ $P = 4.2$ *Exp.*: $P = 4.8$

Reactions in Solution

$$A + B \rightarrow P$$



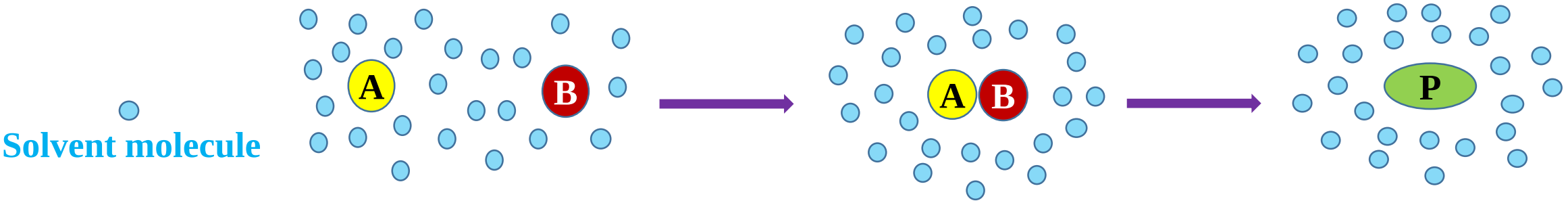
In solution

- **Encounter slowly**
- **Stay near each other longer;**
- **Encounter pair accumulate energy from surrounding**

Cage effect: The lingering of one molecule near another on account of the hindering presence of solvent molecules.

Reactions in Solution

$$A + B \rightarrow P$$



In solution

$$A + B \rightarrow AB \qquad v = k_d[A][B]$$

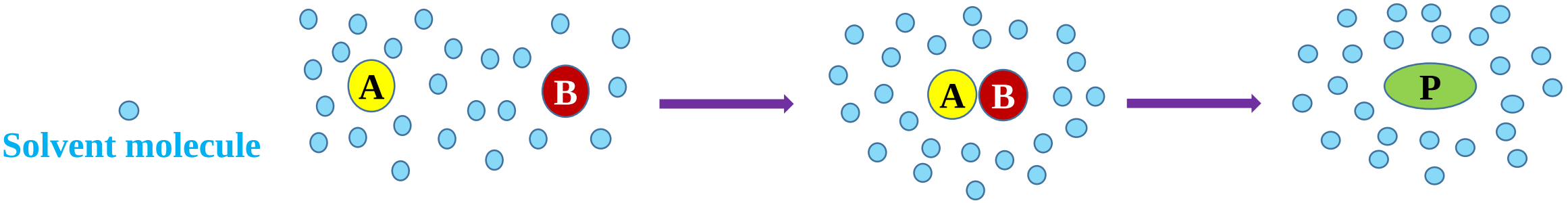
$$AB \rightarrow A + B \qquad v = k'_d[AB]$$

$$AB \rightarrow P \qquad v = k_a[AB]$$

$$A + B \rightarrow P \qquad v = k_2[A][B]$$

Reactions in Solution

$$A + B \rightarrow P$$



In solution

$$A + B \rightarrow AB \qquad v = k_d[A][B]$$

$$AB \rightarrow A + B \qquad v = k'_d[AB]$$

$$AB \rightarrow P \qquad v = k_a[AB]$$

$$A + B \rightarrow P \qquad v = k_2[A][B]$$

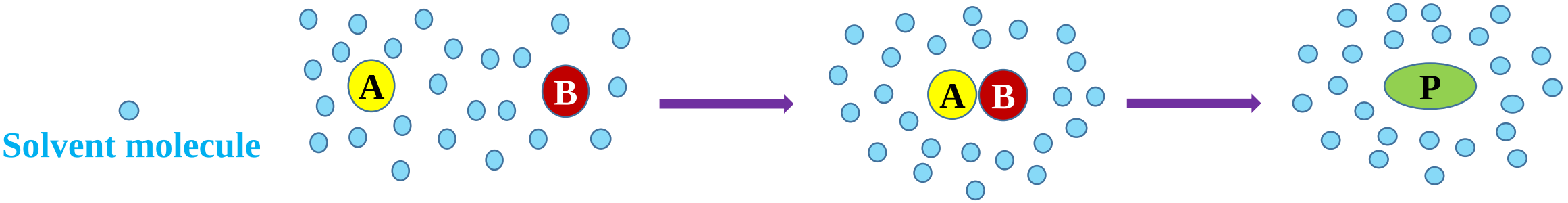
In steady-state approximation:

$$\frac{d[AB]}{dt} = k_d[A][B] - k'_d[AB] - k_a[AB] \approx 0 \rightarrow [AB] = \frac{k_d[A][B]}{k_a + k'_d}$$

$$\frac{d[P]}{dt} \approx k_a[AB] = k_2[A][B] \rightarrow k_2 = \frac{k_a k_d}{k_a + k'_d}$$

Reactions in Solution

$$A + B \rightarrow P$$



In solution

$$\begin{aligned} &A + B \rightarrow AB && v = k_d[A][B] \\ &AB \rightarrow A + B && v = k'_d[AB] \\ &AB \rightarrow P && v = k_a[AB] \\ &A + B \rightarrow P && v = k_2[A][B] \end{aligned}$$

In steady-state approximation:

$$\frac{d[AB]}{dt} = k_d[A][B] - k'_d[AB] - k_a[AB] \approx 0 \rightarrow [AB] = \frac{k_d[A][B]}{k_a + k'_d}$$

$$\frac{d[P]}{dt} \approx k_a[AB] = k_2[A][B] \rightarrow k_2 = \frac{k_a k_d}{k_a + k'_d}$$

Diffusion-control ● $k'_d \ll k_a \rightarrow k_2 \approx \frac{k_a k_d}{k_a} = k_d$

$k_d = \frac{8RT}{3\eta}$ **> ~ 10^9 dm^3 mol^{-1} s^{-1}**

η: viscosity

Activation-control ● $k_a \ll k'_d \rightarrow k_2 \approx \frac{k_a k_d}{k'_d} = k_a K$

K is equilibrium constant for A+B ↔ AB

Estimating a diffusional rate constant

$$k_d = \frac{8RT}{3\eta}$$

The rate constant for the recombination of I atoms in hexane at 298 K, when the viscosity of the solvent is 0.326 cP (with 1 P = 10^{-1} kg m^{-1} s^{-1}) is

$$k_d = \frac{8 \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})}{3 \times (3.26 \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1})} = 2.0 \times 10^7 \text{ m}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

(100 cP =1 P)

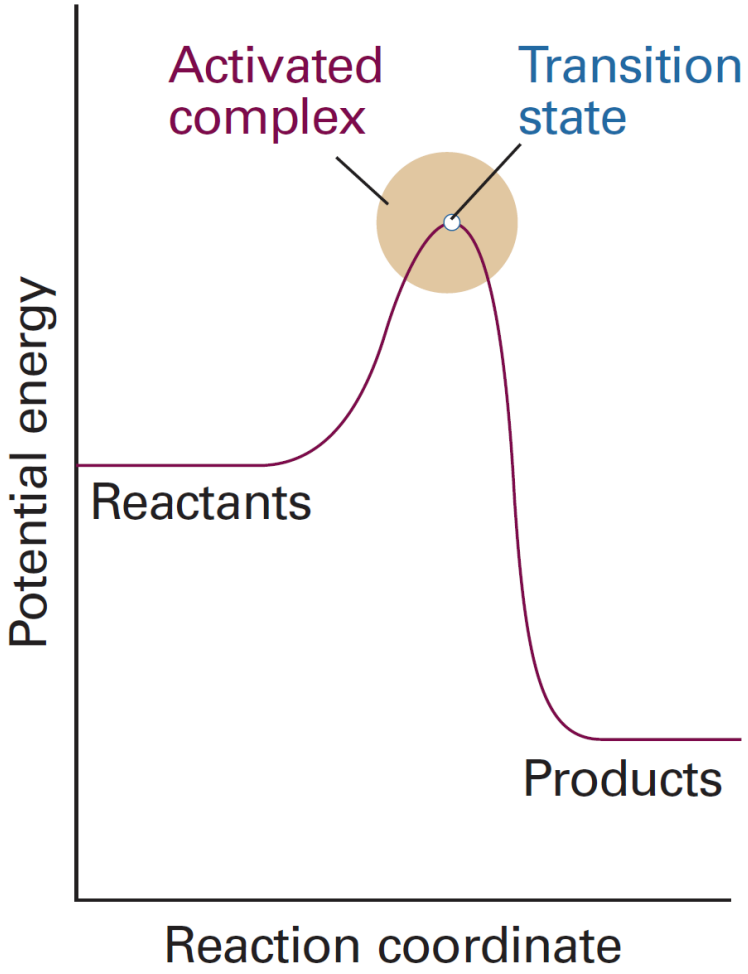
where we have used 1 J = 1 kg m^2 s^{-2}. Because 1 m^3 = 10^3 dm^3, this result corresponds to 2.0×10^{10} dm^3 mol^{-1} s^{-1}. The experimental value is 1.3×10^{10} dm^3 mol^{-1} s^{-1}, so the agreement is very good considering the approximations involved.

Transition State Theory

An approach to estimate the rate constant of
an activation-controlled reaction in both gas-phase and solution

Transition State Theory

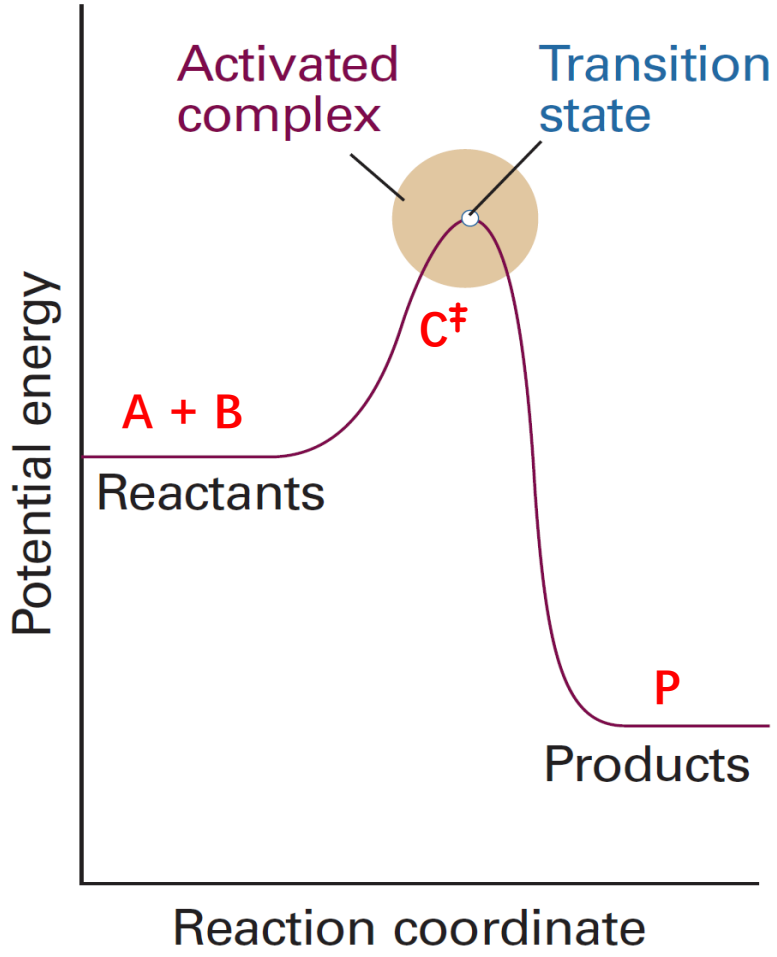
(Activated complex theory)



TST method offers a more detailed calculation of rate constants by **using** the concepts of **statistical thermodynamics**.

Don't need to worry about the steric factor *P* anymore!

The Eyring Equation



In a rapid pre-equilibrium

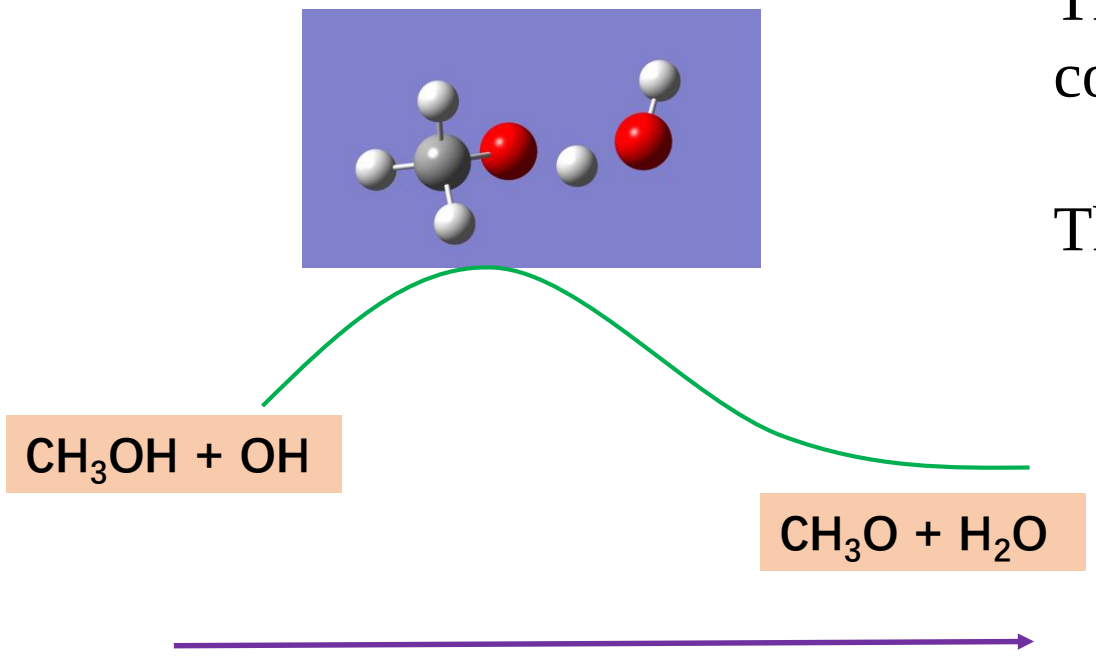
$$A + B \rightleftharpoons C^{\ddagger} \qquad K^{\ddagger} = \frac{p_{C^{\ddagger}} p^{\ominus}}{p_A p_B} \qquad p_J = RT[J] \rightarrow [C^{\ddagger}] = \frac{RT}{p^{\ominus}} K^{\ddagger}[A][B]$$

$$C^{\ddagger} \rightarrow P \qquad v = k^{\ddagger}[C^{\ddagger}] \qquad v = k_2[A][B] \rightarrow k_2 = \frac{RT}{p^{\ominus}} k^{\ddagger} K^{\ddagger}$$

The rate of decay of the activated complex

How to calculate $k^{\ddagger}$ and $K^{\ddagger}$??

$C^{\ddagger} \rightarrow P$ $\quad v = k^{\ddagger}[C^{\ddagger}]$ **?? $k^{\ddagger}$**



The vibrational frequency along the reaction coordinate: ***v***;

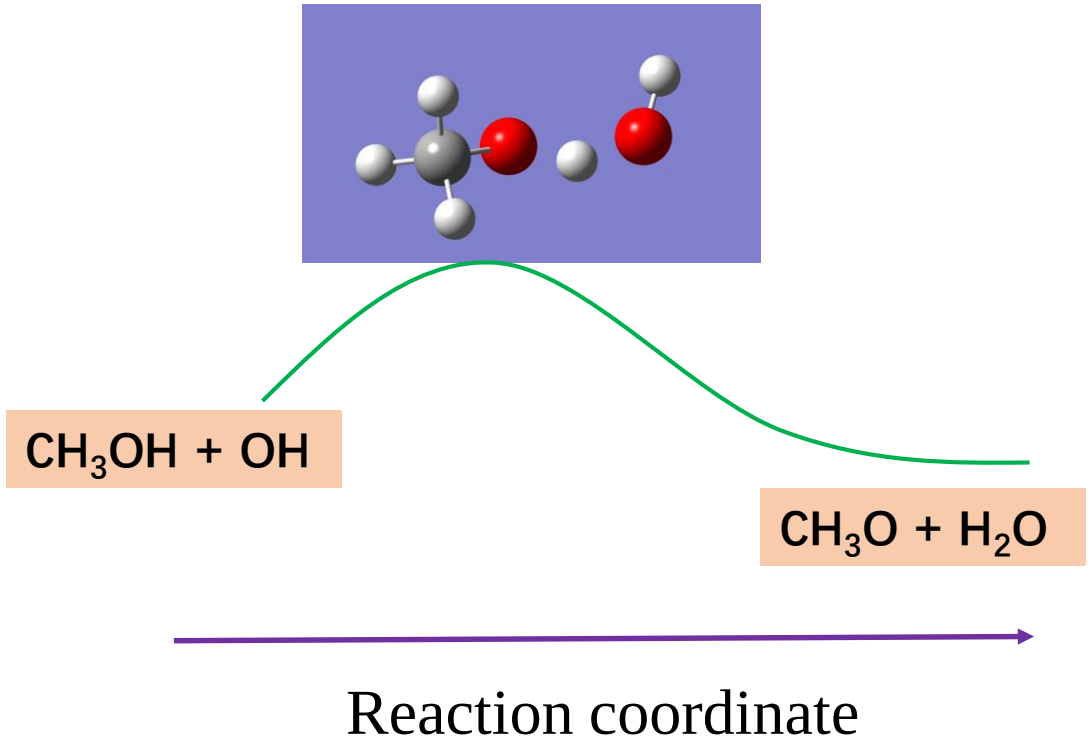
The frequency passing through the transition state: ***v***.

$$k^{\ddagger} = \kappa\nu$$

Transmission coefficient

$$A + B \overset{K^{\ddagger}}{\rightleftharpoons} C^{\ddagger}$$

?? $K^{\ddagger}$



$$K^{\ddagger} = \frac{N_A q^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0 / RT}$$

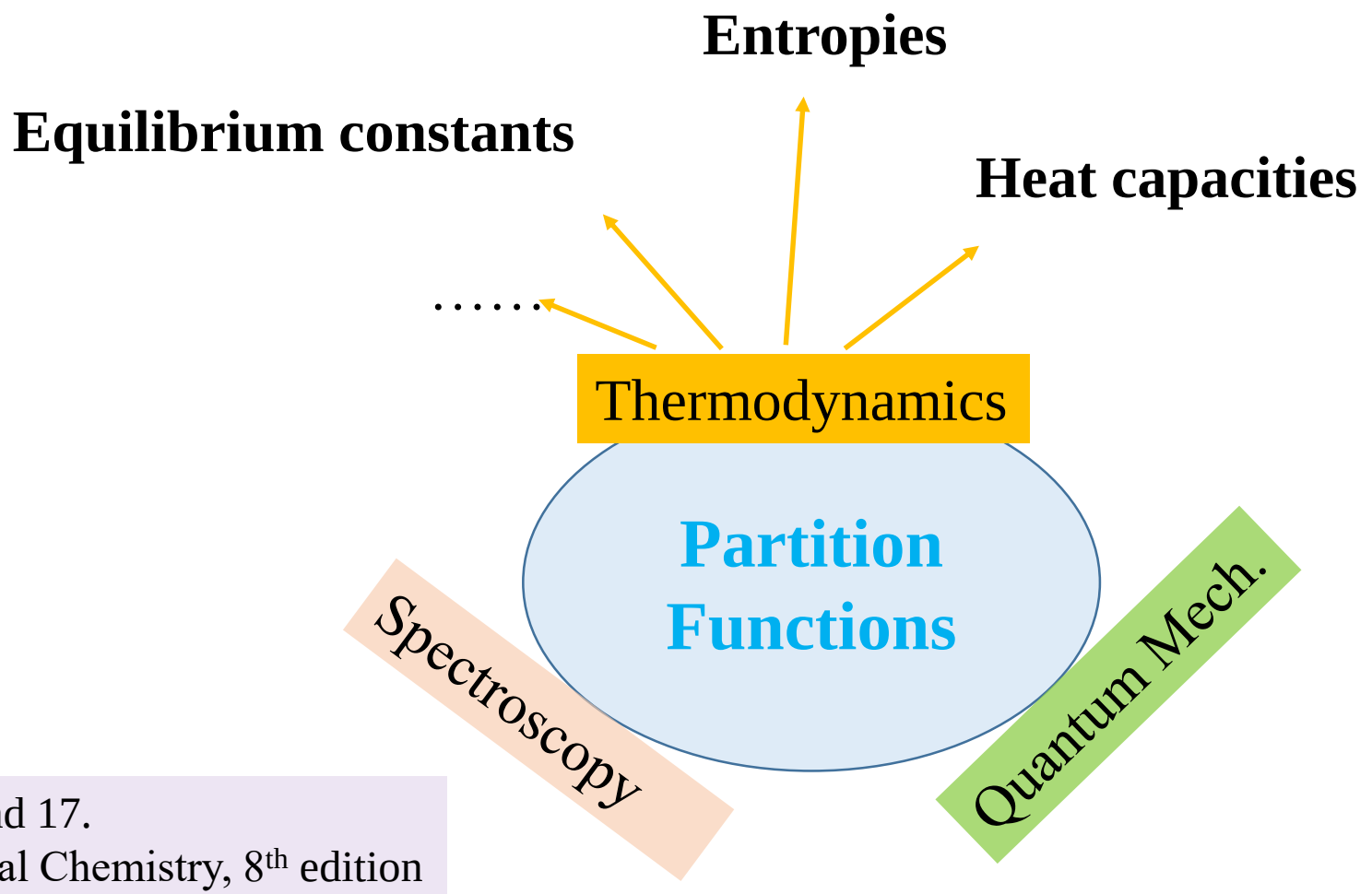
$$\Delta E_0 = E_0(C^{\ddagger}) - E_0(A) - E_0(B)$$

$q^{\ominus}_J$:

The standard **molar partition functions**

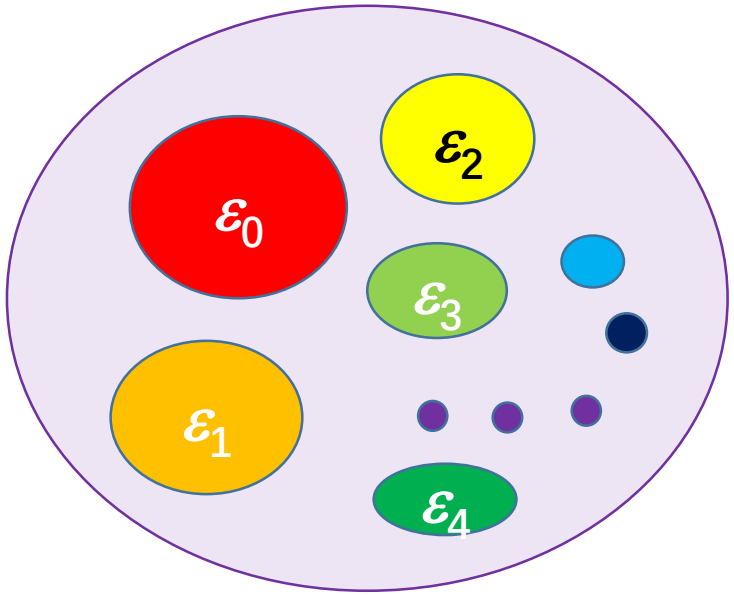
$$p^{\ominus} = 1 \text{ bar}$$

Partition Functions

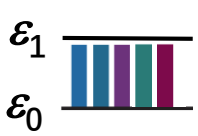


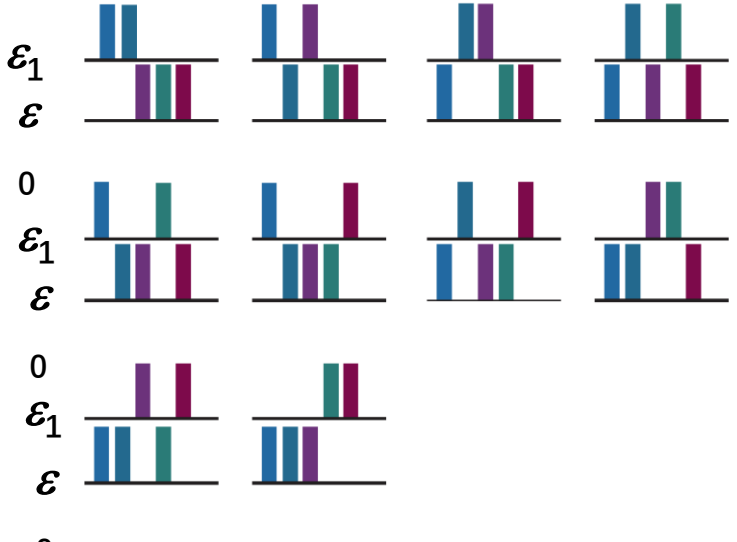
Chapters 16 and 17.
Atkins' Physical Chemistry, 8th edition

Configuration $\{n_0, n_1, \ldots\}$ can be achieved in W ways



1 way for $\{5,0,0,\ldots\}$, **10 ways for** $\{3,2,0,\ldots\}$





n_i molecules in the state with energy ε_i

$$\varepsilon_0 \le \varepsilon_1 \le \varepsilon_2 \ldots .$$

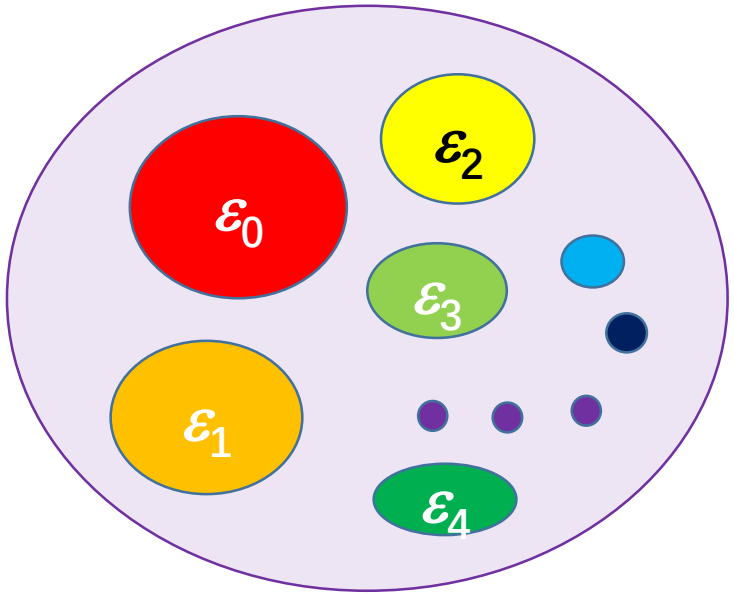
$$\sum_i n_i = N \qquad \sum_i n_i \varepsilon_i = E$$

Weight of $\{n_0, n_1, \ldots\}$ **:**

$$W = \frac{N!}{n_0! n_1! n_2! \ldots}$$

***N, E* are constants**

configuration $\{n_0, n_1, \ldots\}$ with a weight W



The properties of the system are characteristic of the dominating configuration with the greatest W.

Populations in the configuration of greatest weight W :

$$\frac{n_i}{N} = \frac{e^{-\beta\varepsilon_i}}{\sum_i e^{-\beta\varepsilon_i}} \qquad \beta = \frac{1}{kT}$$

n_i molecules in the state with energy ε_i

$$\varepsilon_0 \le \varepsilon_1 \le \varepsilon_2 \ldots .$$

$$\sum_i n_i = N \qquad \boxed{\sum_i n_i \varepsilon_i = E}$$

The **fraction of molecules** in the state i *with energy* ε_i

***N, E* are constants**

The Molecular Partition Function

Boltzmann distribution $p_i = \dfrac{e^{-\beta\varepsilon_i}}{q}$ $p_i = n_i/N$

***q* is the molecular partition function**

$$q = \sum_i e^{-\beta\varepsilon_i} \quad \text{or} \quad q = \sum_{\text{levels } i} g_i e^{-\beta\varepsilon_i}$$

g_i states have the same energy ε_i

$$\lim_{T\to 0} q = g_0 \qquad \lim_{T\to\infty} q = \infty$$

The molecular partition function gives an indication of the number of states that are thermally accessible to a molecule at the temperature of the system.

The Molar Partition Function

$$N = nN_A$$

$$q_m = q/n$$

in mol^{-1}

The Molecular Partition Function

$$\varepsilon_i = \varepsilon_i^{\mathrm{T}} + \varepsilon_i^{\mathrm{R}} + \varepsilon_i^{\mathrm{V}} + \varepsilon_i^{\mathrm{E}}$$

$$q = \sum_i e^{-\beta\varepsilon_i} = \sum_{i\,(\text{all states})} e^{-\beta\varepsilon_i^{\mathrm{T}} - \beta\varepsilon_i^{\mathrm{R}} - \beta\varepsilon_i^{\mathrm{V}} - \beta\varepsilon_i^{\mathrm{E}}}$$

$$= \sum_{i\,(\text{translational})} \sum_{i\,(\text{rotational})} \sum_{i\,(\text{vibrational})} \sum_{i\,(\text{electronic})} e^{-\beta\varepsilon_i^{\mathrm{T}} - \beta\varepsilon_i^{\mathrm{R}} - \beta\varepsilon_i^{\mathrm{V}} - \beta\varepsilon_i^{\mathrm{E}}}$$

$$= \left(\sum_{i\,(\text{translational})} e^{-\beta\varepsilon_i^{\mathrm{T}}} \right) \left(\sum_{i\,(\text{rotational})} e^{-\beta\varepsilon_i^{\mathrm{R}}} \right) \left(\sum_{i\,(\text{vibrational})} e^{-\beta\varepsilon_i^{\mathrm{V}}} \right) \left(\sum_{i\,(\text{electronic})} e^{-\beta\varepsilon_i^{\mathrm{E}}} \right)$$

$$= q^{\mathrm{T}} q^{\mathrm{R}} q^{\mathrm{V}} q^{\mathrm{E}}$$

The translational contribution

$$q^{\mathrm{T}} = \frac{V}{\Lambda^3} \qquad \Lambda = h\left(\frac{\beta}{2\pi m}\right)^{1/2} = \frac{h}{(2\pi mkT)^{1/2}}$$

The rotational contribution

Linear rotors: $q^{\mathrm{R}} = \dfrac{kT}{hcB}$

Nonlinear rotors: $q^{\mathrm{R}} = \left(\dfrac{kT}{hc}\right)^{3/2}\left(\dfrac{\pi}{ABC}\right)^{1/2}$

A, B, and C are the rotational constants of the molecule

The vibrational contribution

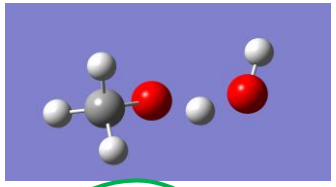
$$q^{\mathrm{V}} = \prod_{i}^{F} \frac{e^{-h\nu_i/2kT}}{1-e^{-h\nu_i/kT}}$$

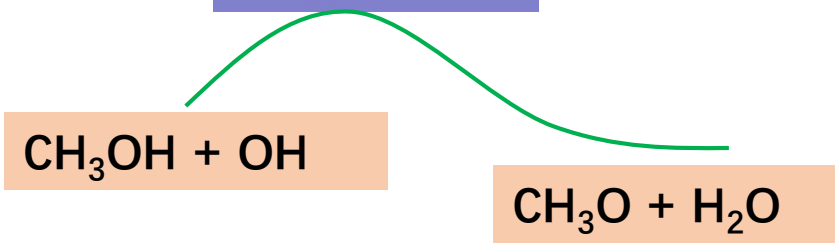
F is the freedom, the number of vibrational modes

The electronic contribution

$$q^{\mathrm{E}} = \sum_{\text{energy levels}} g_j \mathrm{e}^{-\beta\varepsilon_j}$$

$$A + B \overset{K^{\ddagger}}{\leftrightarrow} C^{\ddagger} \overset{k^{\ddagger}}{\rightarrow} P$$



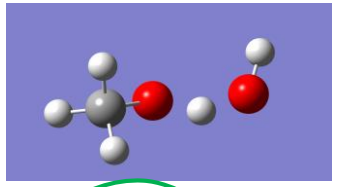


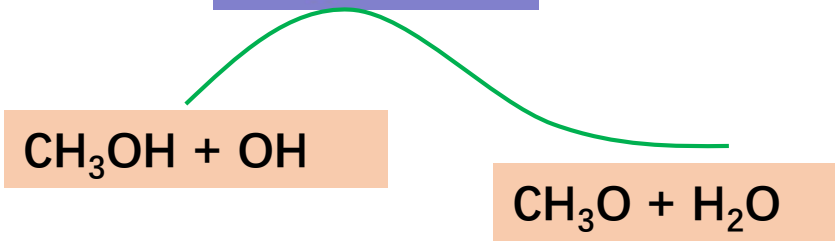
$$v = k_2[\mathrm{A}][\mathrm{B}] \qquad k_2 = \frac{RT}{p^{\ominus}} k^{\ddagger} K^{\ddagger}$$

A vibration of $C^{\ddagger}$ tips it through the TS to product has frequency v

$$k^{\ddagger} = \kappa \nu \qquad K^{\ddagger} = \frac{N_{\mathrm{A}} q^{\ominus}_{\mathrm{C}^{\ddagger}}}{q^{\ominus}_{\mathrm{A}} q^{\ominus}_{\mathrm{B}}} \mathrm{e}^{-\Delta E_0 / RT} \qquad \Delta E_0 = E_0(\mathrm{C}^{\ddagger}) - E_0(\mathrm{A}) - E_0(\mathrm{B})$$

$$A + B \overset{K^{\ddagger}}{\leftrightarrow} C^{\ddagger} \overset{k^{\ddagger}}{\rightarrow} P$$





$$v = k_2[A][B] \qquad k_2 = \frac{RT}{p^{\ominus}} k^{\ddagger} K^{\ddagger}$$

A vibration of C‡ tips it through the TS to product has frequency v

$$k^{\ddagger} = \kappa\nu \qquad K^{\ddagger} = \frac{N_A q^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT} \qquad \Delta E_0 = E_0(C^{\ddagger}) - E_0(A) - E_0(B)$$

The partition function for the vibration of C‡ with freq. v

$$q = \frac{1}{1 - e^{-h\nu/kT}}$$

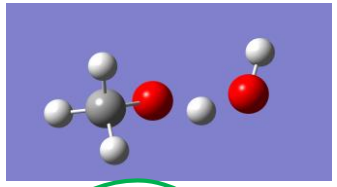
v is much lower, so $h\nu/kT \ll 1$

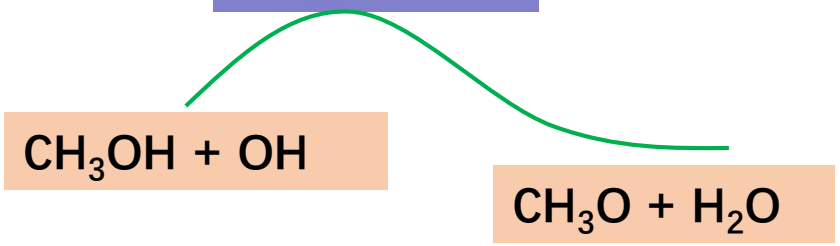
$$\rightarrow \quad q = \frac{1}{1 - \left(1 - \frac{h\nu}{kT} + \cdots\right)} \approx \frac{kT}{h\nu}$$

$$q_{C^{\ddagger}} \approx \frac{kT}{h\nu} \bar{q}_{C^{\ddagger}} \quad \rightarrow \quad K^{\ddagger} = \frac{kT}{h\nu} \bar{K}^{\ddagger} \qquad \bar{K}^{\ddagger} = \frac{N_A \bar{q}^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT}$$

The partition function for all the other modes of C‡

$$A + B \overset{K^{\ddagger}}{\leftrightarrow} C^{\ddagger} \overset{k^{\ddagger}}{\rightarrow} P$$





$$v = k_2[\mathrm{A}][\mathrm{B}] \qquad k_2 = \frac{RT}{p^{\ominus}} k^{\ddagger} K^{\ddagger}$$

A vibration of $C^{\ddagger}$ tips it through the TS to product has frequency v

$$k^{\ddagger} = \kappa\nu \qquad K^{\ddagger} = \frac{N_A q^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT} \qquad \Delta E_0 = E_0(C^{\ddagger}) - E_0(A) - E_0(B)$$

The partition function for the vibration of $C^{\ddagger}$ with freq. v

$$q = \frac{1}{1 - e^{-h\nu/kT}}$$

v is much lower, so $h\nu/kT \ll 1$

$$\rightarrow \quad q = \frac{1}{1 - \left(1 - \frac{h\nu}{kT} + \cdots\right)} \approx \frac{kT}{h\nu}$$

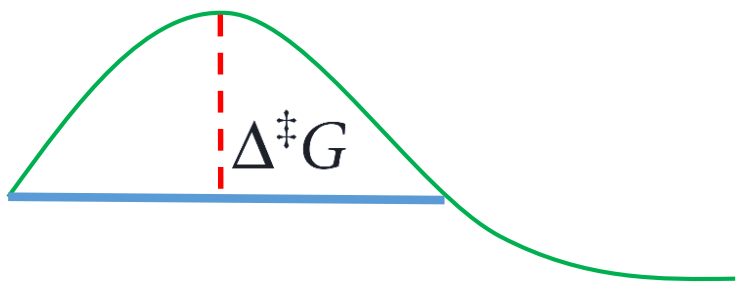
$$q_{C^{\ddagger}} \approx \frac{kT}{h\nu} \bar{q}_{C^{\ddagger}} \quad \rightarrow \quad K^{\ddagger} = \frac{kT}{h\nu} \bar{K}^{\ddagger} \qquad \bar{K}^{\ddagger} = \frac{N_A \bar{q}^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT}$$

The partition function for all the other modes of $C^{\ddagger}$

$$\rightarrow \quad k_2 = k^{\ddagger} \frac{RT}{p^{\ominus}} K^{\ddagger} = \kappa\nu \frac{kT}{h\nu} \frac{RT}{p^{\ominus}} \bar{K}^{\ddagger} = \kappa \frac{kT}{h} \bar{K}^{\ddagger}_c$$

Eyring equation

$$A + B \overset{K^{\ddagger}}{\leftrightarrow} C^{\ddagger} \overset{k^{\ddagger}}{\rightarrow} P$$



$$v = k_2[A][B]$$

$$k_2 = k^{\ddagger}\frac{RT}{p^{\ominus}}K^{\ddagger} = \kappa\nu\frac{kT}{h\nu}\frac{RT}{p^{\ominus}}\bar{K}^{\ddagger}$$

Gibbs energy of activation

$$\Delta^{\ddagger}G = -RT\ln\bar{K}^{\ddagger} \qquad \text{(Standard)}$$

Eyring equation

$$k_2 = \kappa\frac{kT}{h}\frac{RT}{p^{\ominus}}e^{-\Delta^{\ddagger}G/RT}$$

Electron Transfer in Homogeneous System

Electron transfer from a donor species D to an acceptor species A in solution.

$$D + A \rightarrow D^+ + A^- \qquad v = k_{obs}[D][A] \qquad K = \frac{[D^+][A^-]}{[D][A]}$$

$$D + A \rightleftharpoons DA \qquad K_{DA} = \frac{k_a}{k'_a} = \frac{[DA]}{[D][A]}$$

$$DA \rightarrow D^+A^- \qquad v_{et} = k_{et}[DA]$$

$$D^+A^- \rightarrow DA \qquad v_r = k_r[D^+A^-]$$

$$D^+A^- \rightarrow D^+ + A^- \qquad v_d = k_d[D^+A^-]$$

$$\frac{1}{k_{obs}} = \frac{1}{k_a} + \frac{k'_a}{k_a k_{et}}\left(1 + \frac{k_r}{k_d}\right)$$

We assume $k_d >> k_r$ ➡ $\frac{1}{k_{obs}} \approx \frac{1}{k_a}\left(1 + \frac{k'_a}{k_{et}}\right)$

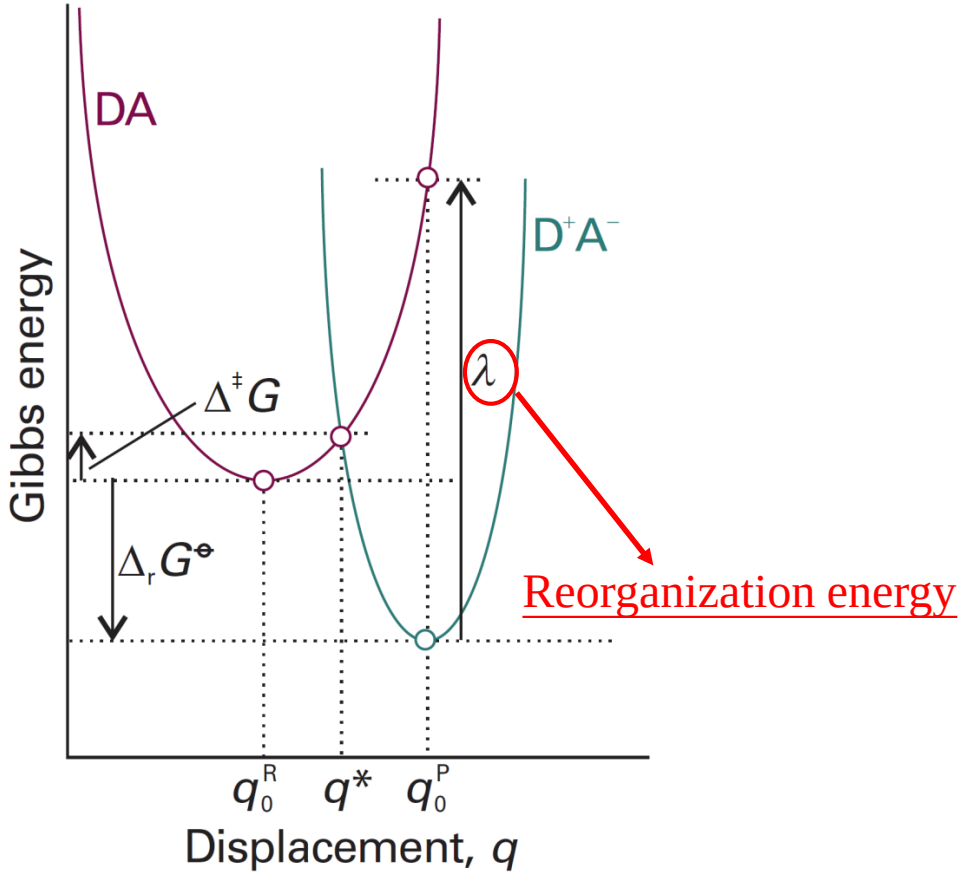
When $k_{et} \gg k'_a$ ➡ $k_{obs} \approx k_a$

When $k_{et} \ll k'_a$ ➡ $k_{obs} \approx (k_a/k'_a)k_{et}$

$$k_{et} = \kappa\nu e^{-\Delta^{\ddagger}G/RT} \qquad k_{obs} \approx K_{DA}k_{et}$$

Theory of Electron Transfer Processes

$$DA \rightarrow D^+A^- \qquad v_{et} = k_{et}[DA] \qquad k_{et} = \kappa\nu e^{-\Delta^{\ddagger}G/RT}$$

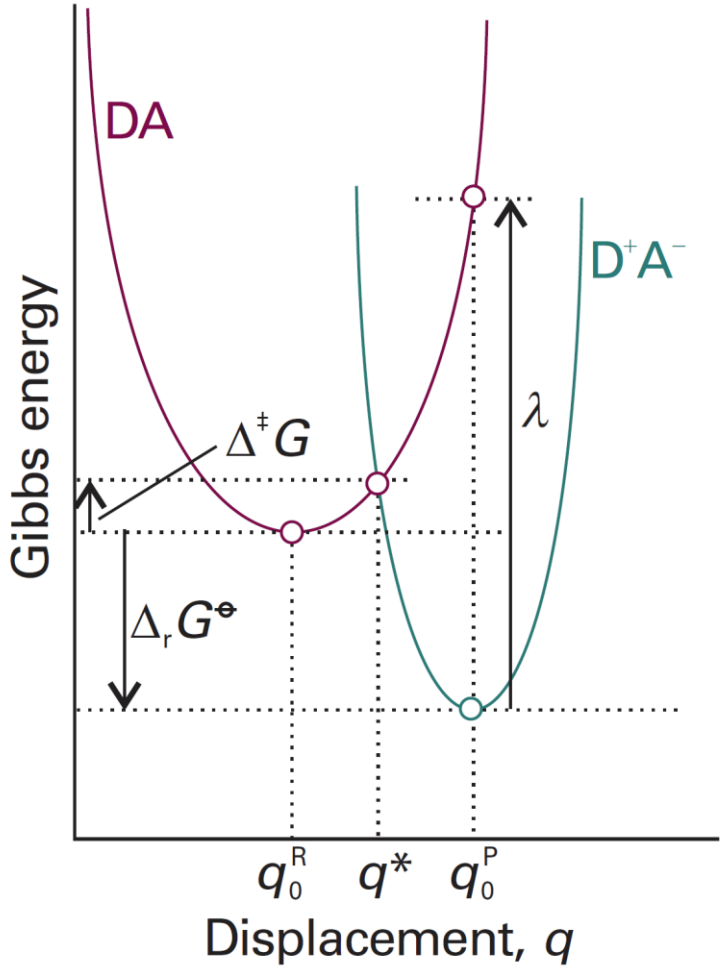


- Electrons are transferred by tunnelling through a potential energy barrier. Electron tunneling influences the magnitude of $\kappa\nu$.
- The complex DA and the solvent molecules surrounding it undergo structural rearrangements prior to electron transfer. The energy associated with these rearrangements and the standard reaction Gibbs energy determine $\Delta^{\ddagger}G$.

------ by R.A. Marcus, N.S. Hush, V.G. Levich, and R.R. Dogonadze.

Electron Tunnelling

$$DA \rightarrow D^+A^- \qquad v_{et} = k_{et}[DA] \qquad k_{et} = \kappa\nu e^{-\Delta^{\ddagger}G/RT}$$

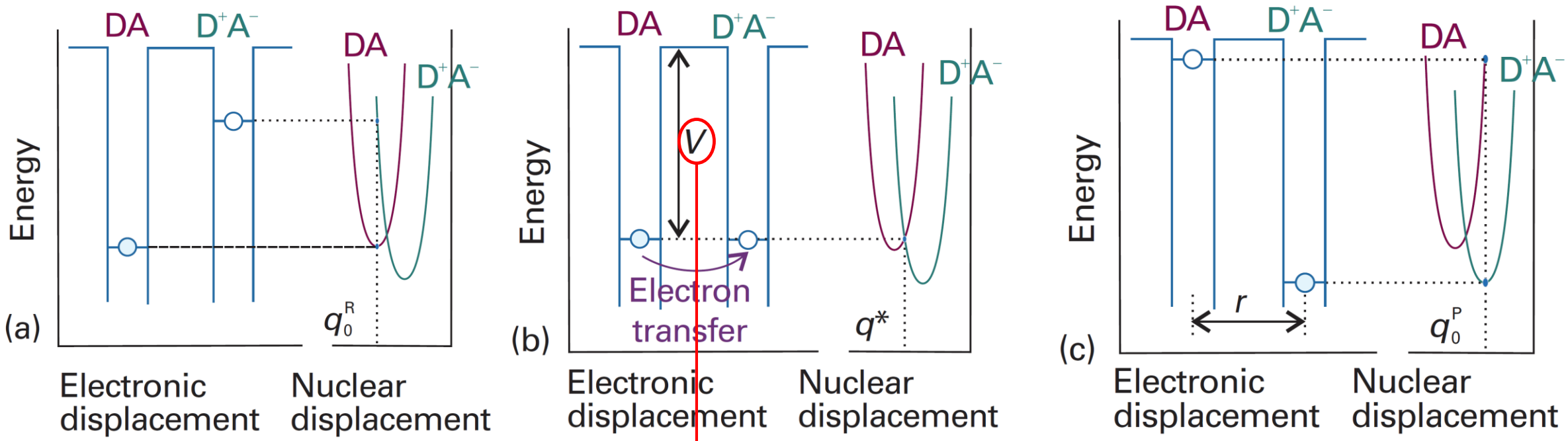


Franck-Condon principle: electronic transitions are so fast that they can be regarded as taking place in a stationary nuclear framework.

→ Electron transfer can occur only after thermal fluctuations bring the geometry of DA to *q**

The factor ***κν* is a measure of the probability** that the system will convert from DA to D^+A^- at *q** by electron transfer within the thermally excited DA complex.

Correspondence between the electronic energy levels and the nuclear energy levels



Increases with the ionization energies of the DA to D^+A^- complexes.

The rate of an electron transition: the exponential dependence on the edge-to-edge distance *r* between D and A

$$\propto \quad \langle H_{DA} \rangle^2 = \langle H^{\circ}_{DA} \rangle^2 e^{-\beta r} \qquad r = 0$$

$$\langle H_{DA} \rangle = \int \psi_A \hat{H}_{DA} \psi_B \, d\tau$$

Electronic coupling matrix element

The rate of electron transfer

$$k_{et} = \frac{2\langle H_{DA}\rangle^2}{h}\left(\frac{\pi^3}{4\lambda RT}\right)^{1/2} e^{-\Delta^{\ddagger}G/RT}$$

Limitations:

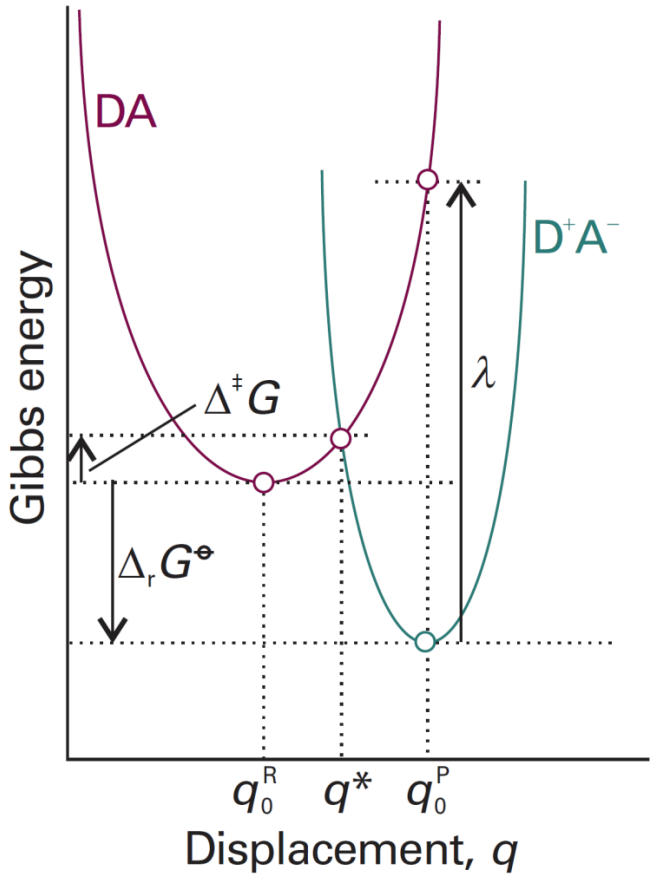
- For weak coupling;
- For high *T*;

$$\Delta^{\ddagger}G = \frac{(\Delta_r G^{\ominus} + \lambda)^2}{4\lambda}$$

Reorganization Energy

- Relative reorientation of D and A in DA
- Relative reorientation of the solvent molecules surrounding DA

Theory of Electron Transfer Processes



$$DA \rightarrow D^+A^- \qquad v_{et} = k_{et}[DA]$$

$$k_{et} = \kappa \nu e^{-\Delta^{\ddagger}G/RT}$$

The k_{et} depends on:

☺the standard reaction Gibbs energy $\Delta_r G^{\ominus}$

☺The edge-to-edge distance between D and A ***r***

The theory has been extended to transfer of other light particles, e.g., protons.

Experimental tests of the dependence of k_{et} **on** ***r***

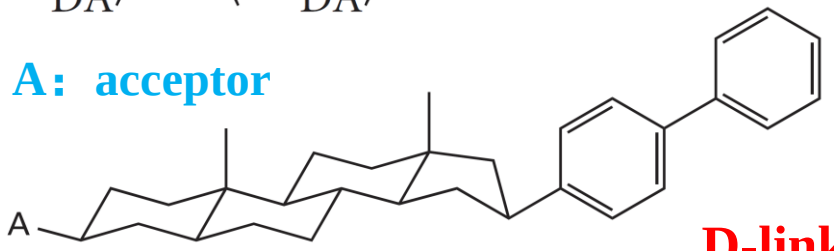
- Difficult when the reactants are ions or molecules free to move in solution.
- Meaningful for systems in which the donor and acceptor are positioned at a variety of distances, by molecular linkers.

$$k_{et} = \frac{2\langle H_{DA}\rangle^2}{h}\left(\frac{\pi^3}{4\lambda RT}\right)^{1/2} e^{-\Delta^{\ddagger}G/RT}$$

$$\langle H_{DA}\rangle^2 = \langle H^{\circ}_{DA}\rangle^2 e^{-\beta r}$$

$$\ln k_{et} = -\beta r + \text{constant}$$

β depends on the medium through which ***e*** must travel from D to A



D-linker-A

Experimental tests of the dependence of k_{et} **on** $\Delta_r G^{\ominus}$

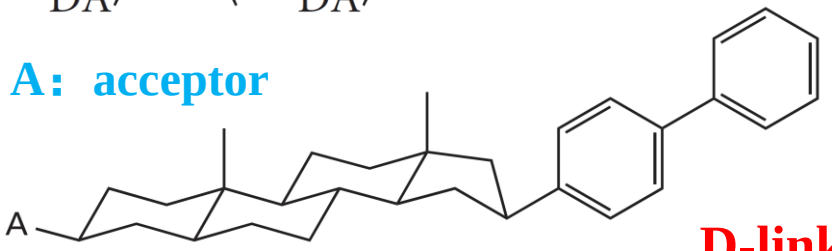
- for systems where r, λ, and $\kappa\nu$ are constants

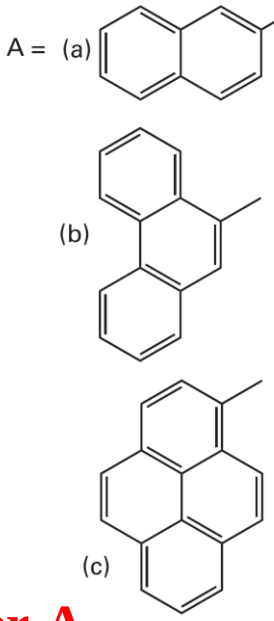
$$\ln k_{et} = -\frac{1}{4\lambda}\left(\frac{\Delta_r G^{\ominus}}{RT}\right)^2 - \frac{1}{2}\left(\frac{\Delta_r G^{\ominus}}{RT}\right) + \text{constant}$$

$$k_{et} = \frac{2\langle H_{DA}\rangle^2}{h}\left(\frac{\pi^3}{4\lambda RT}\right)^{1/2} e^{-\Delta^{\ddagger}G/RT}$$

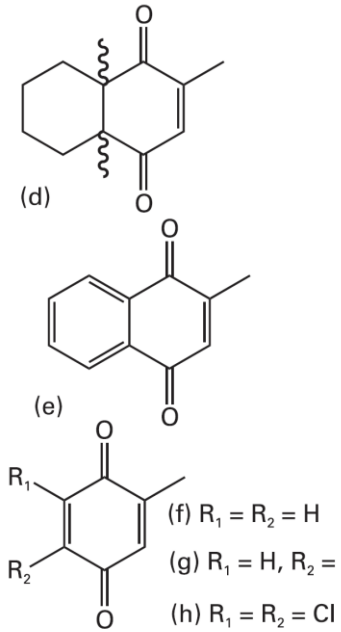
$$\langle H_{DA}\rangle^2 = \langle H^{\circ}_{DA}\rangle^2 e^{-\beta r}$$

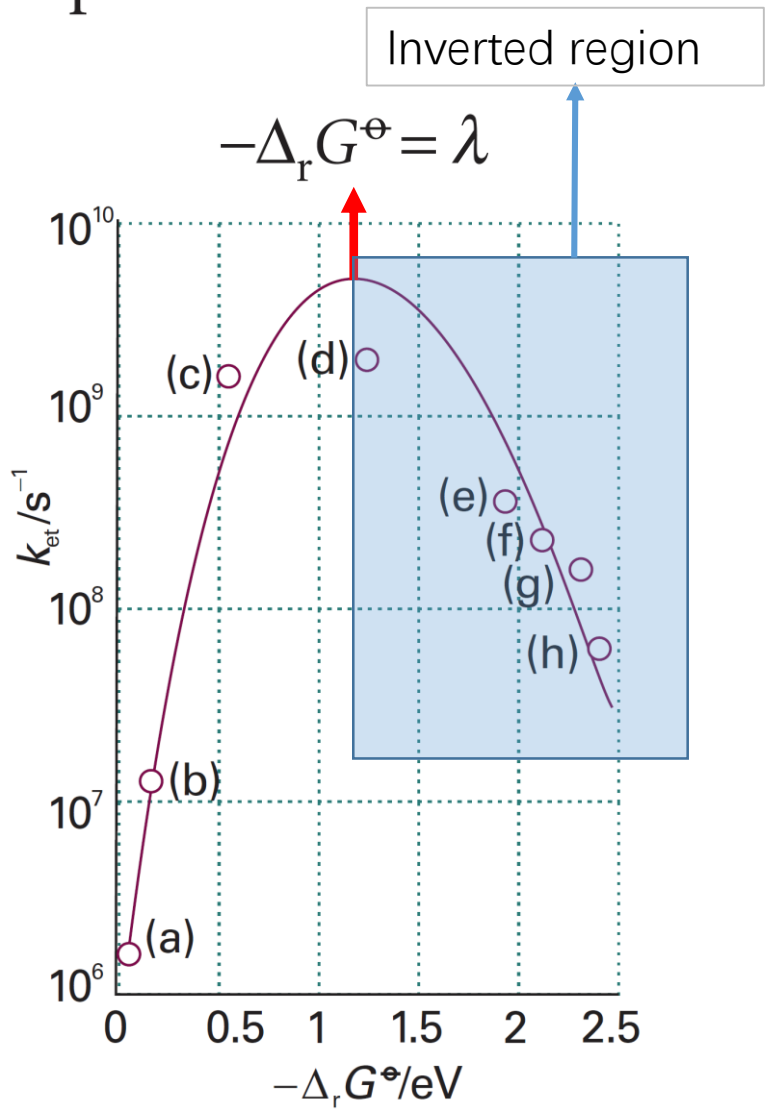
A：acceptor





D-linker-A



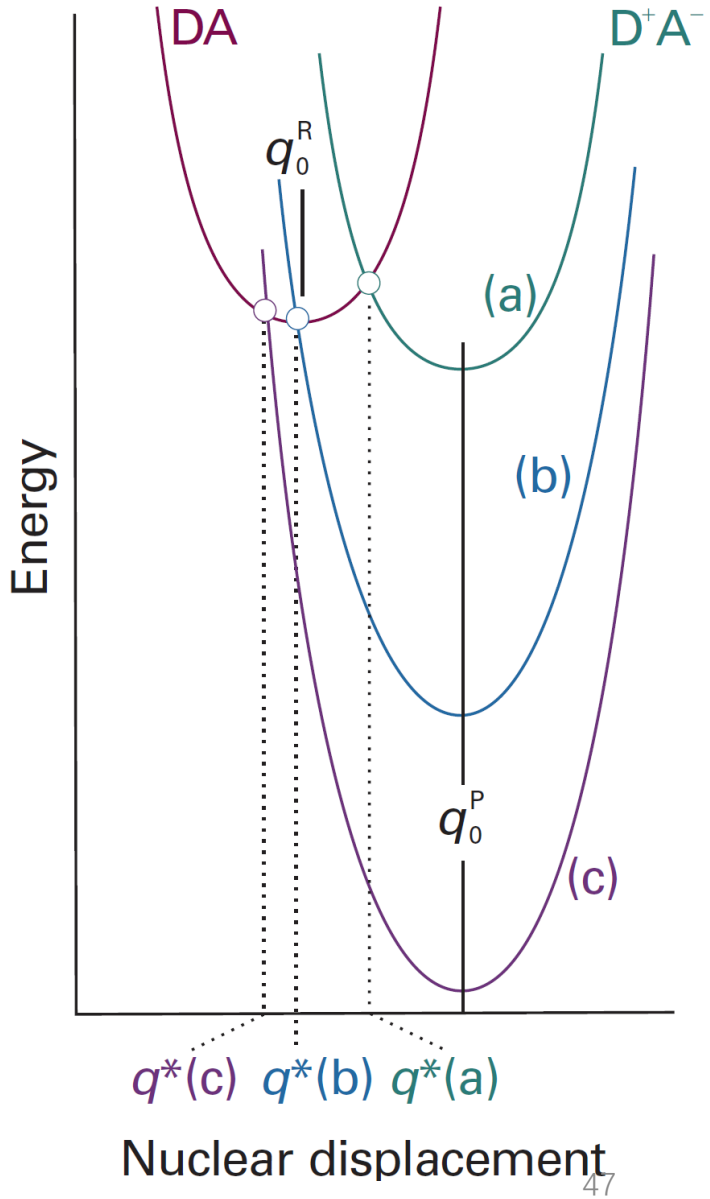


(a) $\Delta^{\ddagger}G > 0$ and $q^*(a) > q^R_0$: process more exergonic, $\Delta^{\ddagger}G$ decreases, and k_{et} increases;

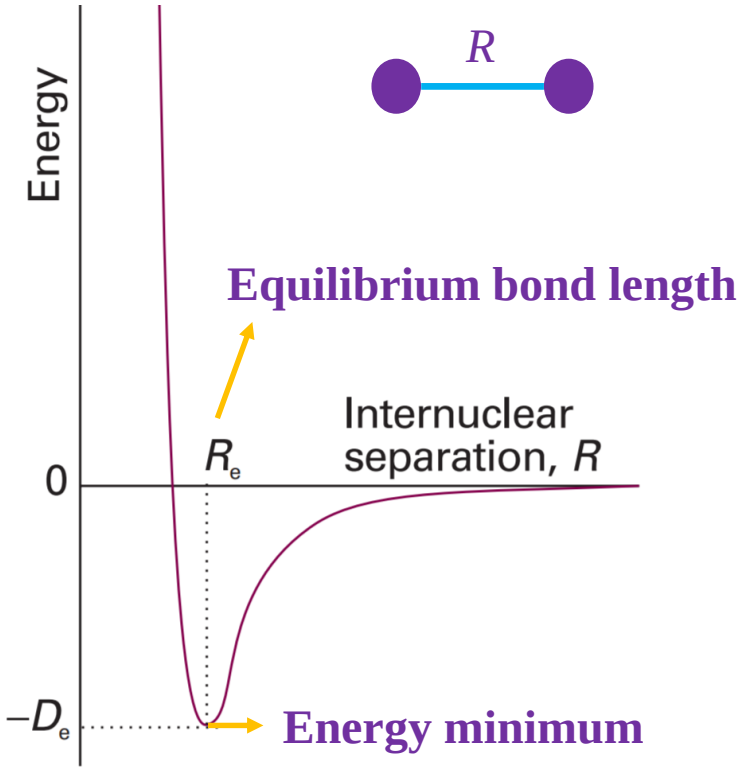
(b) $\Delta^{\ddagger}G = 0$ and $q^*(b) = q^R_0$: k_{et} reaches a maximum;

(c) Process even more exergonic, $\Delta^{\ddagger}G > 0$ again, $q^*(c) < q^R_0$: $\Delta_r G^{\ominus}$ decrease (more negative), $\Delta^{\ddagger}G$ increases, and k_{et} decreases.

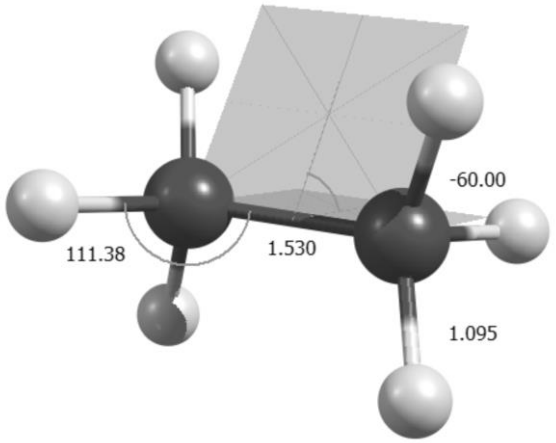
(c) is a good explanation for "inverted region"!



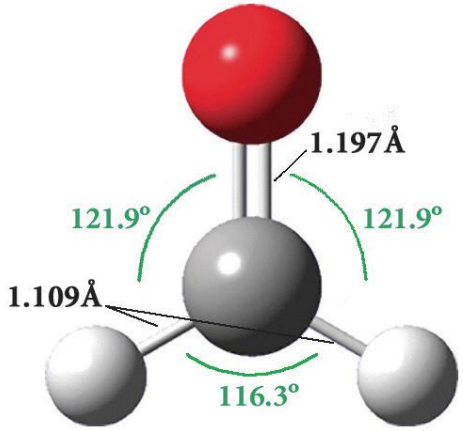
The Potential Energy Surface



Molecular geometry

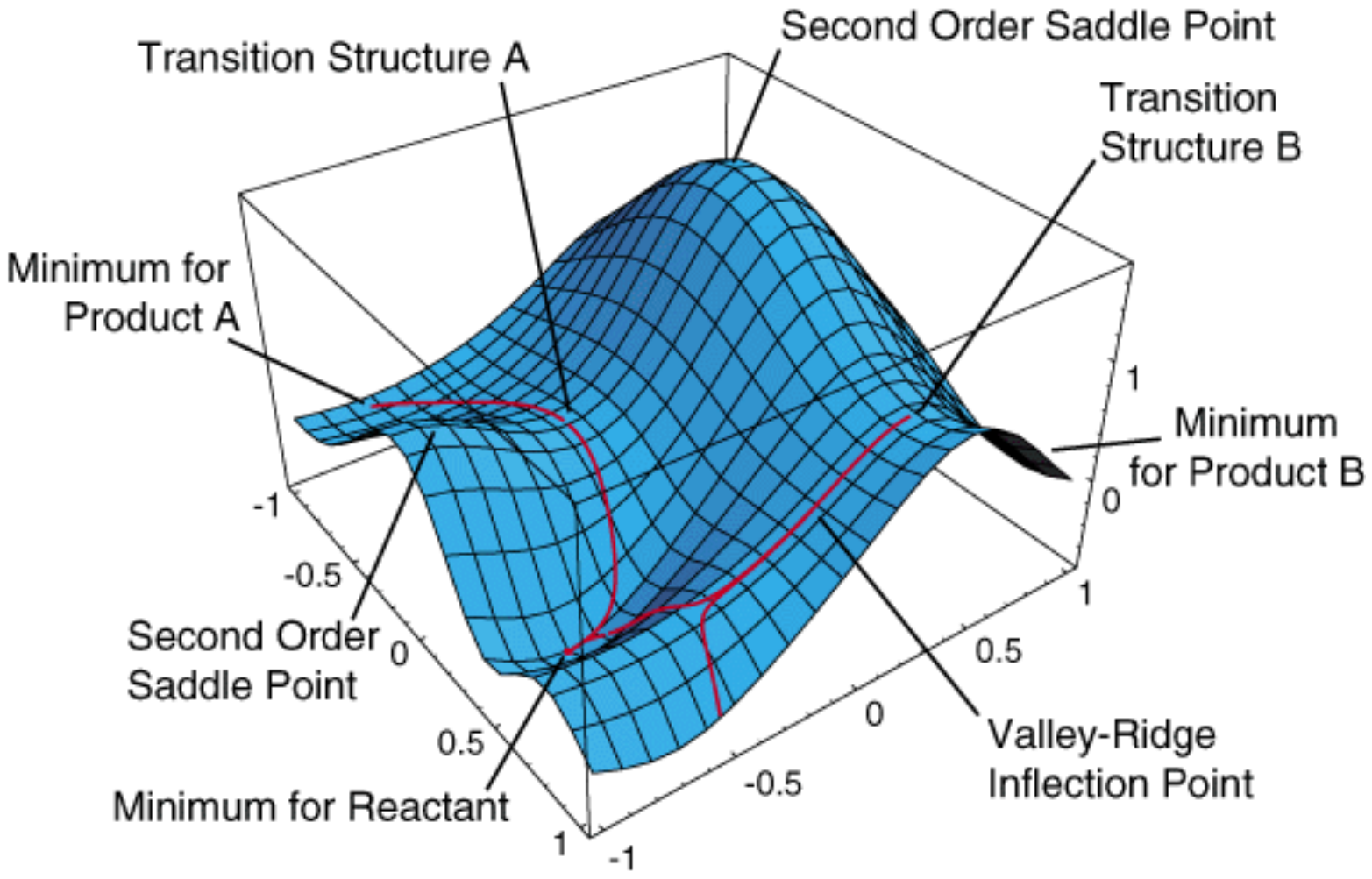


Ethane

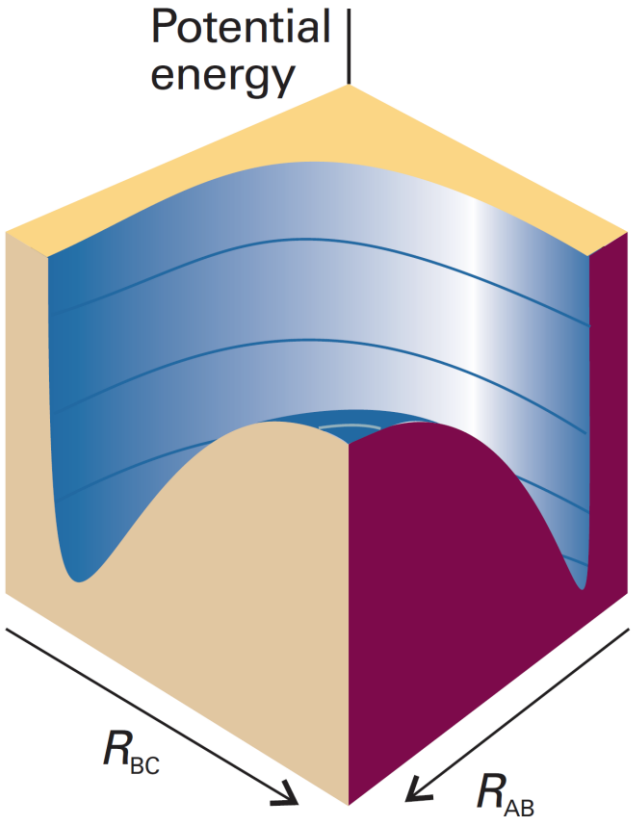


Formaldehyde

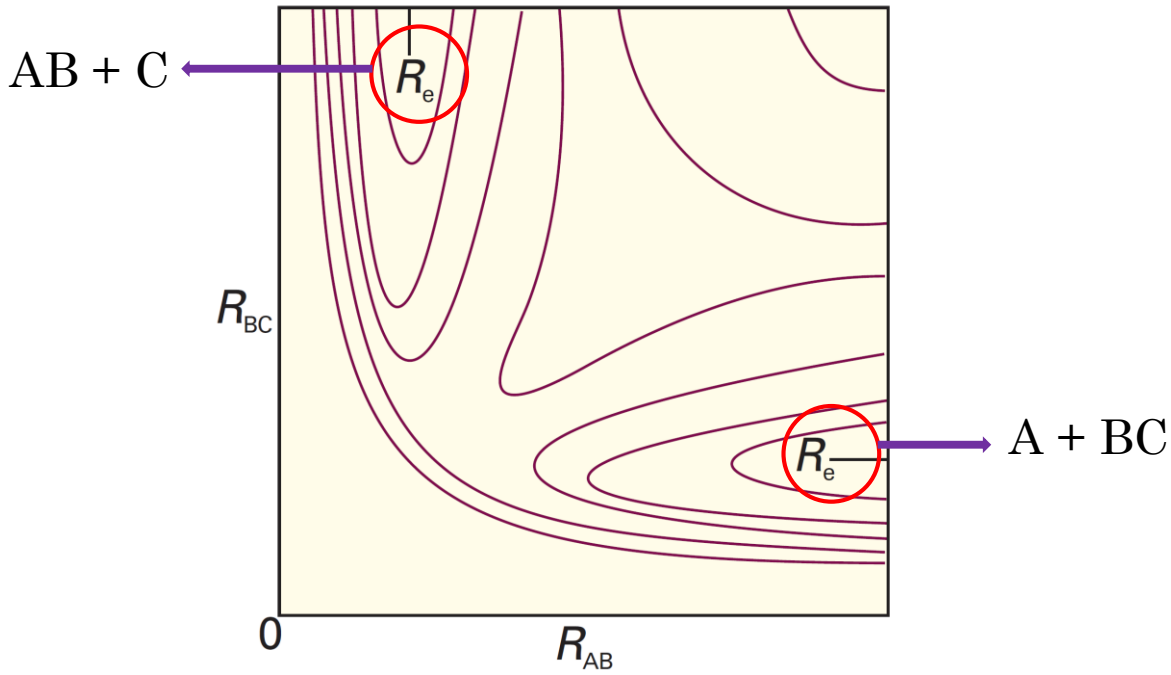
The Potential Energy Surface



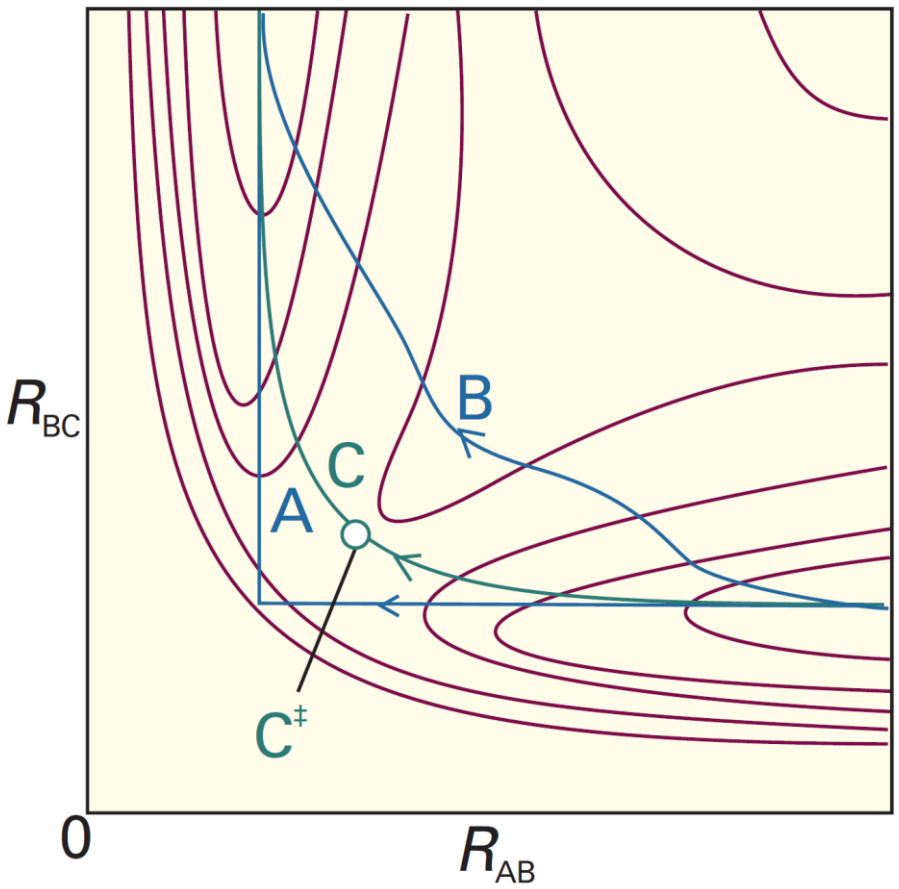
A + BC → AB + C



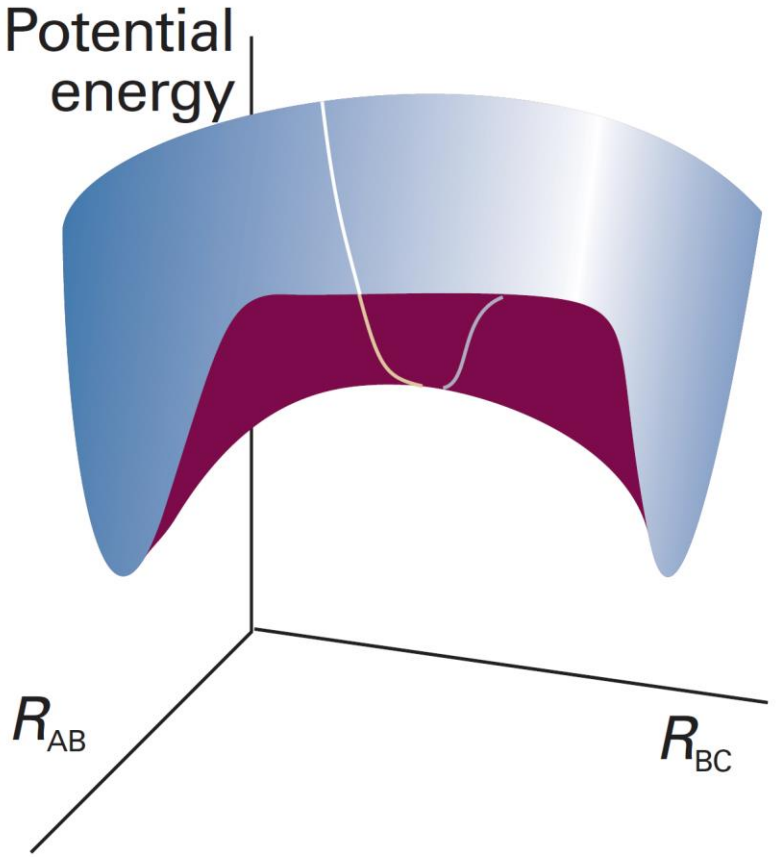
The contour diagram (with contours of equal potential energy)



Collinear approach: two geometrical parameters R_{AB} and R_{BC}



Various trajectories through the potential energy surface



The transition state is a set of configurations (here, marked by the line across the saddle point) through which successful reactive trajectories must pass.

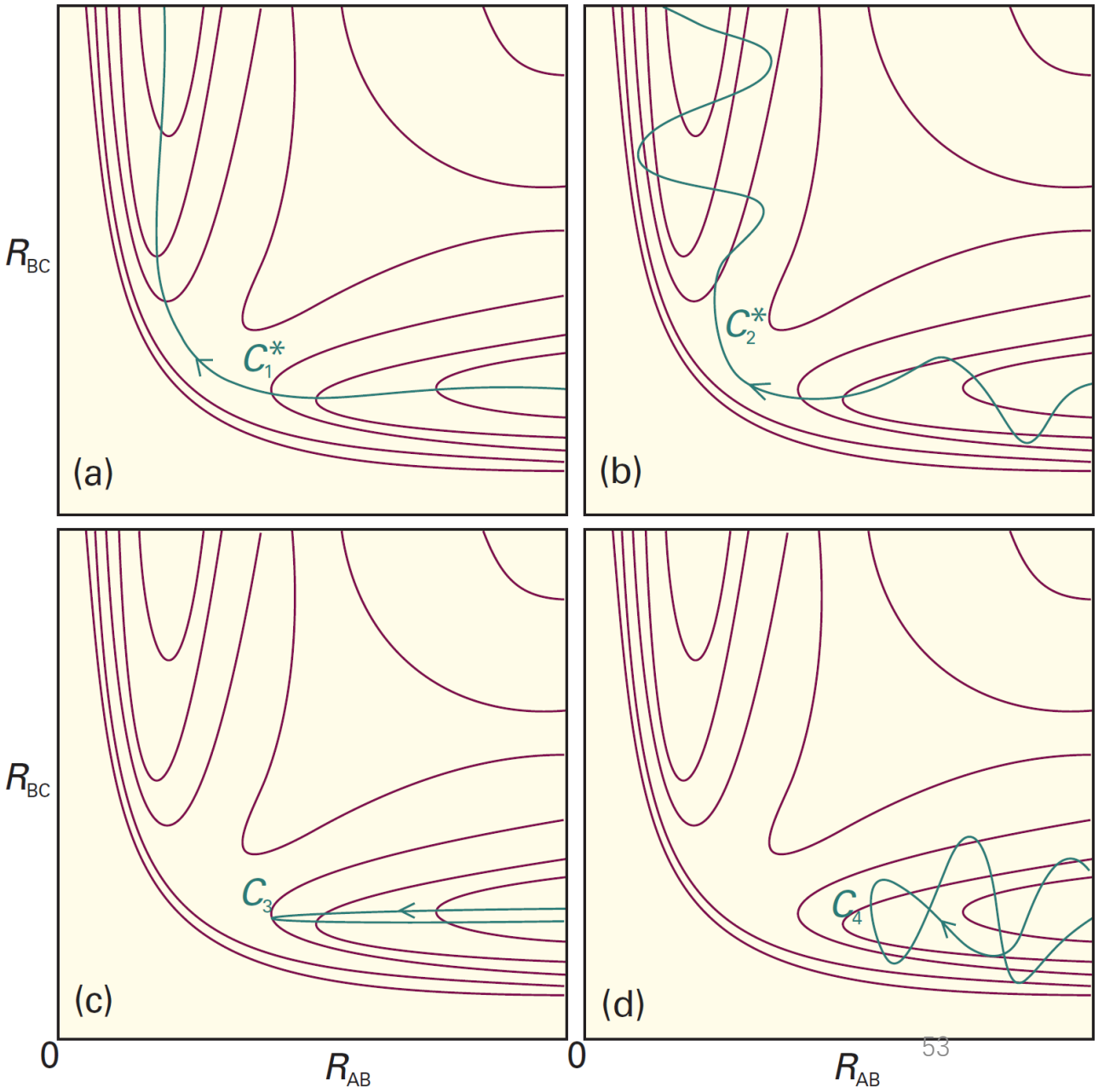
Some successful (*) and unsuccessful encounters.

(a) C_1* : the path along the foot of the valley.

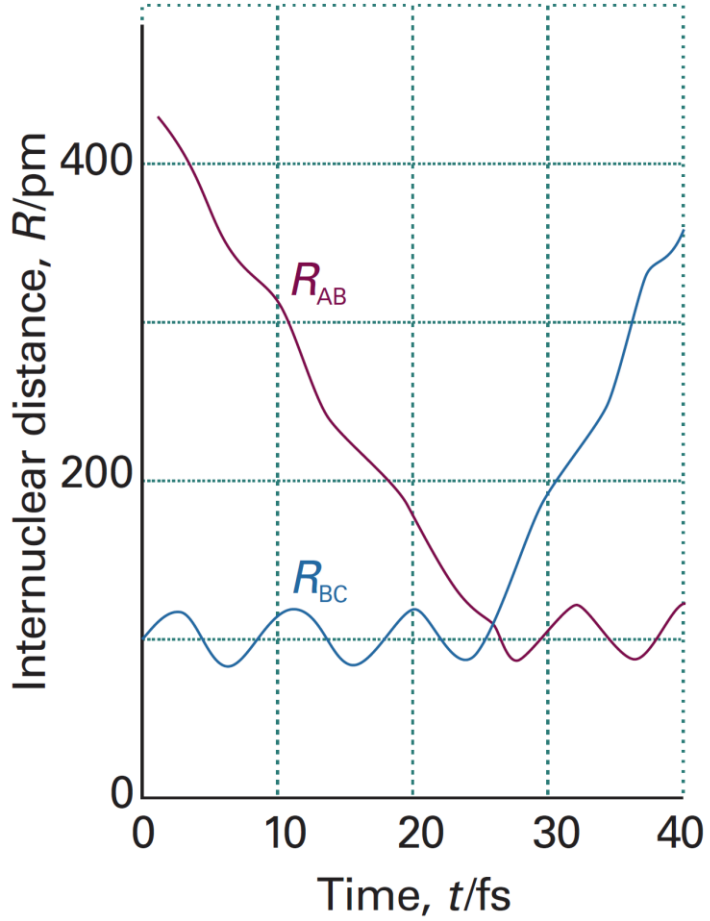
(b) C_2* : an approach of A to a vibrating BC molecule, and the formation of a vibrating AB molecule as C departs.

(c) C_3 : A approaching a non-vibrating BC molecule, but with insufficient translational kinetic energy.

(d) C_4 : A approaching a vibrating BC molecule, but still the energy, and the phase of the vibration, is insufficient for reaction.



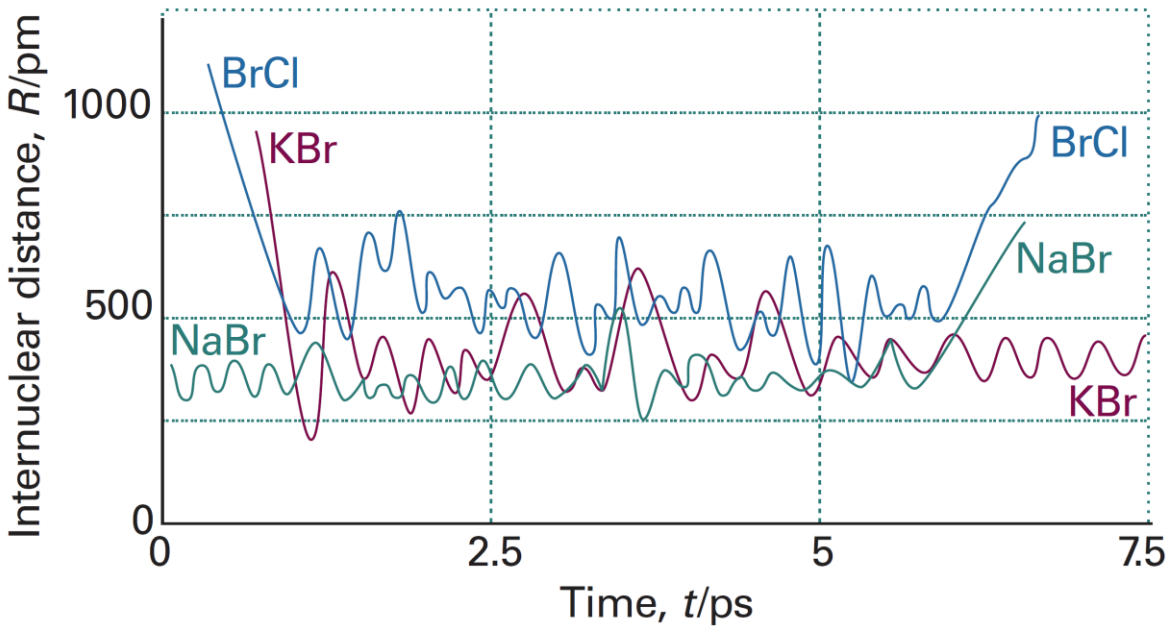
Classical Trajectories



$$H + H_2 \rightarrow H_2 + H$$

A direct mode process, very rapidly

Use classical mechanics to calculate the trajectories of the atoms taking place in a reaction from a set of **initial conditions**, such as velocities, relative orientations, and internal energies of the reacting particles.



$$KCl + NaBr \rightarrow KBr + NaCl$$

A complex mode process, takes a longer time

Reaction Dynamics vs. Chemical Kinetics

- Study **why** chemical reactions occur, **how** to predict their behavior, and **how** to control them.
- It is closely related to chemical kinetics, but is concerned with **individual chemical events** on **atomic length scales** and over very **brief time periods**.
- It considers **state-to-state kinetics** between reactant and product molecules in specific quantum states, and **how energy is distributed** between translational, vibrational, rotational, and electronic modes.

- Or reaction kinetics
- Study **rates** of chemical processes.
- Study **how** different experimental **conditions can influence the speed** of a chemical reaction
- **Yield information** about the reaction's mechanism and transition states
- **Construct mathematical models** that can describe the characteristics of a chemical reaction.

------ from Wikipedia

Checklist of key ideas I

- In **collision theory**: the **rate** $\boldsymbol{k} \propto$ the collision frequency, a steric factor, and the fraction of collisions with at least kinetic energy E_a along their lines of centres.
- A reaction in solution may be a **diffusion-controlled** or **activation-controlled** reaction.
- In **transition state theory**: an activated complex is in equilibrium with the reactants; the rate at which that complex forms products depends on the rate at which it passes through a transition state. The result is the **Eyring equation.**

Checklist of key ideas II

- The **rate constant of electron transfer** in a donor–acceptor complex depends on the distance between electron donor and acceptor, the standard reaction Gibbs energy, and the reorganization energy, λ.
- A **potential energy surface** maps the potential energy as a function of the relative positions of all the atoms taking part in a reaction.

Calculate the magnitude of the diffusion-controlled rate constant at 298 K for the recombination of two atoms in benzene, for which η = 0.601 cP. Assuming the concentration of the reacting species is 1.8 mmol dm^{-3} initially, how long does it take for the concentration of the atoms to fall to half that value? Assume the reaction is elementary.

The diffusion-controlled rate constant is

$$k_d = \frac{8RT}{3\eta} = \frac{8 \times (8.3145\,\mathrm{J\,K^{-1}\,mol^{-1}}) \times (298\,\mathrm{K})}{3 \times (0.601 \times 10^{-3}\,\mathrm{kg\,m^{-1}\,s^{-1}})} = \boxed{1.10 \times 10^{7}\,\mathrm{m^3\,mol^{-1}\,s^{-1}}}$$

The recombination reaction has a rate of

$$v = k_d[\mathrm{A}][\mathrm{B}] \quad \text{with } [\mathrm{A}] = [\mathrm{B}]$$

The half-life is given by

$$t_{1/2} = \frac{1}{k[\mathrm{A}]_0} = \frac{1}{(1.10 \times 10^{10}\,\mathrm{L\,mol^{-1}\,s^{-1}}) \times (1.8 \times 10^{-3}\,\mathrm{mol\,L^{-1}})} = \boxed{5.05 \times 10^{-8}\,\mathrm{s}}$$

(a) Use the collision theory of gas-phase reactions to calculate the theoretical pre-exponential factor A at 350 K for the reaction Cl (g) + H_2 (g) → HCl (g) + H (g), and estimate the steric factor *P* by comparing your calculate A to the experimental observation (lg[A/(mol^{-1}dm^{3}s^{-1})]=10.08). (m_{Cl}=35.45 gmol^{-1},m_{H2}=2.016 gmol^{-1}, d_{Cl} =0.20 nm, d_{H2}=0.15 nm);

(b) Take the partition function for each freedom of motion as an approximate value, for example, q_t≈10^{10}, q_r≈10, q_v≈1, and take the electronic contribution to the partition function to be 1 (q_e=1), please prove the transition state of the reaction in (a) to be a non-linear molecule. (hint: $q=q^Tq^Rq^Vq^E$)

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The theoretical pre-exponential factor

$$A = \sigma \left(\frac{8k_B T}{\pi\mu}\right)^{\frac{1}{2}} N_A$$

$$= \pi\left(\frac{1}{2}(d_{Cl} + d_{H_2})\right)^2 \left(\frac{8k_B T}{\pi\left(\frac{m_{Cl} m_{H_2}}{m_{Cl}+m_{H_2}}\right)}\right)^{\frac{1}{2}} N_A = 1.14 \times 10^{11} \ \text{mol}^{-1}\text{dm}^3\text{s}^{-1}$$

$$P = \frac{10^{10.08}}{1.14 \times 10^{11}} = 0.105$$

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$$k_2 = \kappa \frac{kT}{h} \bar{K}_c^{\ddagger}$$

$$\bar{K}_c^{\ddagger} = (RT/p^{\ominus})\bar{K}^{\ddagger}$$

$$\bar{K}^{\ddagger} = \frac{N_A \bar{q}_{C^{\ddagger}}^{\ominus}}{q_A^{\ominus} q_B^{\ominus}} e^{-\Delta E_0/RT}$$

$q_{Cl}^{\ominus} = (q_{Cl}^{T})^3 = 10^{30}$

$q_{H_2}^{\ominus} = (q_{H_2}^{T})^3 (q_{H_2}^{R})^2 q_{H_2}^{V} = 10^{32}$

For linear TS, $q_{TS^{\neq}}^{\ominus} = (q_{TS}^{T})^3 (q_{TS}^{R})^2 (q_{TS}^{V})^3 = 10^{32}$ $A = \frac{(RT)^2}{hp^{\ominus}} \frac{q_{TS^{\neq}}^{o}}{q_{Cl}^{\ominus} q_{H_2}^{o}} = 1.28 \times 10^{35} \times 10^{-30} =$

1.28×10^5 mol^{-1}m^{3}s^{-1} = 1.28×10^8 mol^{-1}dm^{3}s^{-1}

For non-linear TS, $q_{TS^{\neq}}^{o} = (q_{TS}^{T})^3 (q_{TS}^{R})^3 (q_{TS}^{V})^2 = 10^{33}$ $A = \frac{(RT)^2}{hp^{\ominus}} \frac{q_{TS^{\neq}}^{o}}{q_{Cl}^{\ominus} q_{H_2}^{o}} = 1.28 \times 10^{35} \times$

$10^{-29} = 1.28 \times 10^6$ mol^{-1}m^{3}s^{-1} = 1.28×10^9 mol^{-1}dm^{3}s^{-1}

The A obtained for non-linear TS is more close to the experimental value, thus the transition state of this reaction may be a non-linear molecule.

Collision Theory

$k_2 \propto$

- Encounter rate
- Minimum energy requirement
- Steric requirement

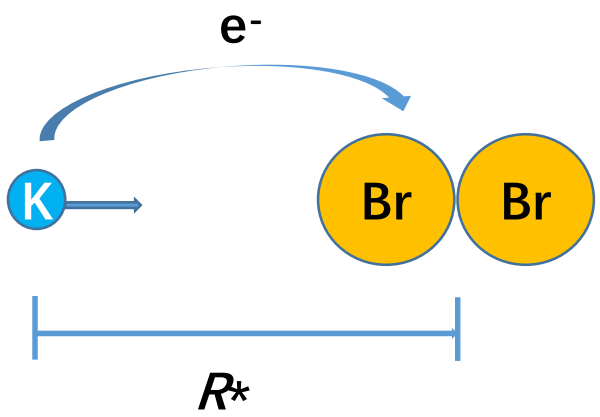
$$A + B \rightarrow P \qquad v = k_2[A][B]$$

$$k_2 = \underbrace{P\sigma}_{\sigma^*}\left(\frac{8kT}{\pi\mu}\right)^{1/2} N_A e^{-E_a/RT}$$

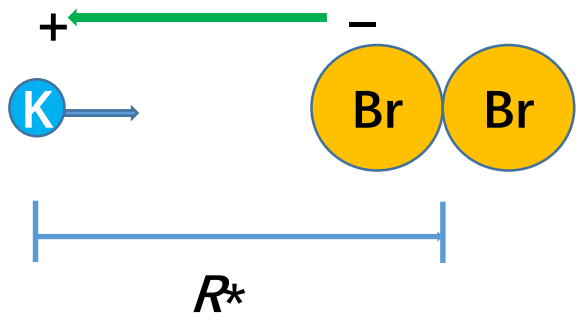
P is normally <<1

Harpoon Reaction

$P > 1$



Coulombic attraction



$$E = I - E_{ea} - \frac{e^2}{4\pi\varepsilon_0 R}$$

$= 0$ (R=R*)

$$P = \frac{\sigma^*}{\sigma} = \frac{R^{*2}}{d^2}$$

Reactions in Solution

$$A + B \rightarrow P \qquad v = k_2[A][B]$$

$$\rightarrow \begin{cases} A + B \rightarrow AB & v = k_d[A][B] \\ AB \rightarrow A + B & v = k'_d[AB] \\ AB \rightarrow P & v = k_a[AB] \end{cases}$$

In steady-state approximation:

$$\frac{d[AB]}{dt} = k_d[A][B] - k'_d[AB] - k_a[AB] \approx 0 \Rightarrow [AB] = \frac{k_d[A][B]}{k_a + k'_d}$$

$$\frac{d[P]}{dt} \approx k_a[AB] = k_2[A][B] \Rightarrow \boxed{k_2 = \frac{k_a k_d}{k_a + k'_d}}$$

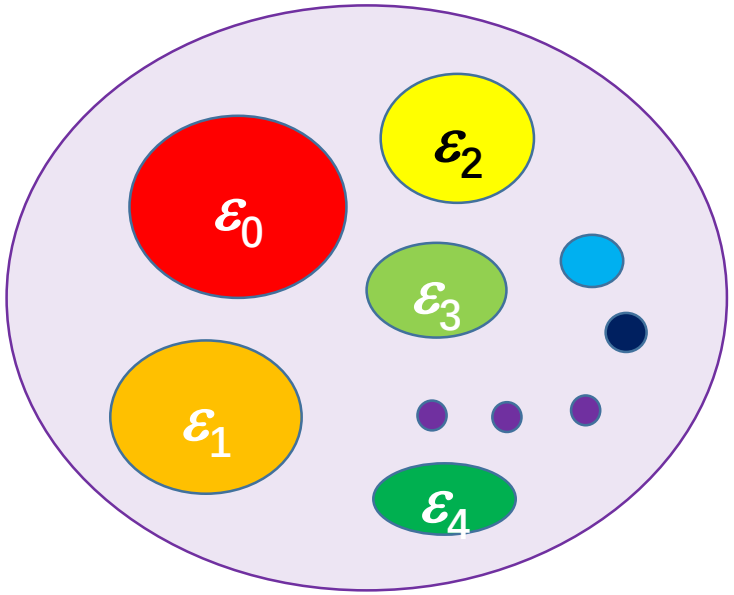
- **Diffusion-control** $k'_d \ll k_a \Rightarrow k_2 \approx \frac{k_a k_d}{k_a} = k_d \qquad k_d = \frac{8RT}{3\eta}$

  η: viscosity

- **Activation-control** $k_a \ll k'_d \Rightarrow k_2 \approx \frac{k_a k_d}{k'_d} = k_a K$

  K is equilibrium constant for A+B ↔ AB

Boltzmann Distribution



$\varepsilon_0 \leq \varepsilon_1 \leq \varepsilon_2 \ldots .$

n_i molecules in the state with energy ε_i

Constant total energy: $\sum_i n_i \varepsilon_i = E$

Constant total number of molecules: $\sum_i n_i = N$

The **fraction of molecules** in the state *i*

$$\frac{n_i}{N} = \frac{e^{-\beta\varepsilon_i}}{\sum_i e^{-\beta\varepsilon_i}}$$

$$\beta = \frac{1}{kT}$$

Boltzmann constant

Molecular Partition Functions

$$q = \sum_i e^{-\beta\varepsilon_i} \quad \text{or} \quad q = \sum_{\text{levels } i} g_i e^{-\beta\varepsilon_i}$$

g_i states have the same energy ε_i

$$\lim_{T\to 0} q = g_0 \qquad \lim_{T\to\infty} q = \infty$$

The molecular partition function gives an indication of the number of states that are thermally accessible to a molecule at the temperature of the system.

Molar partition function $q_m = q/n$ $\qquad N = nN_A$

Molecular Partition Functions

$$\varepsilon_i = \varepsilon_i^{\mathrm{T}} + \varepsilon_i^{\mathrm{R}} + \varepsilon_i^{\mathrm{V}} + \varepsilon_i^{\mathrm{E}}$$

$$q = \sum_i e^{-\beta\varepsilon_i} = \sum_{i\,(\text{all states})} e^{-\beta\varepsilon_i^{\mathrm{T}} - \beta\varepsilon_i^{\mathrm{R}} - \beta\varepsilon_i^{\mathrm{V}} - \beta\varepsilon_i^{\mathrm{E}}}$$

$$= \sum_{i\,(\text{translational})} \sum_{i\,(\text{rotational})} \sum_{i\,(\text{vibrational})} \sum_{i\,(\text{electronic})} e^{-\beta\varepsilon_i^{\mathrm{T}} - \beta\varepsilon_i^{\mathrm{R}} - \beta\varepsilon_i^{\mathrm{V}} - \beta\varepsilon_i^{\mathrm{E}}}$$

$$= \left(\sum_{i\,(\text{translational})} e^{-\beta\varepsilon_i^{\mathrm{T}}} \right) \left(\sum_{i\,(\text{rotational})} e^{-\beta\varepsilon_i^{\mathrm{R}}} \right) \left(\sum_{i\,(\text{vibrational})} e^{-\beta\varepsilon_i^{\mathrm{V}}} \right) \left(\sum_{i\,(\text{electronic})} e^{-\beta\varepsilon_i^{\mathrm{E}}} \right)$$

$$= q^{\mathrm{T}} q^{\mathrm{R}} q^{\mathrm{V}} q^{\mathrm{E}}$$

The translational contribution

$$q^{\mathrm{T}} = \frac{V}{\Lambda^3} \qquad \Lambda = h\left(\frac{\beta}{2\pi m}\right)^{1/2} = \frac{h}{(2\pi m k T)^{1/2}}$$

The rotational contribution

Linear rotors: $q^{\mathrm{R}} = \dfrac{kT}{hcB}$ Nonlinear rotors: $q^{\mathrm{R}} = \left(\dfrac{kT}{hc}\right)^{3/2}\left(\dfrac{\pi}{ABC}\right)^{1/2}$

A, B, and C are the rotational constants of the molecule

The vibrational contribution

$$q^{\mathrm{V}} = \prod_{i}^{F} \frac{e^{-h\nu_i/2kT}}{1-e^{-h\nu_i/kT}}$$

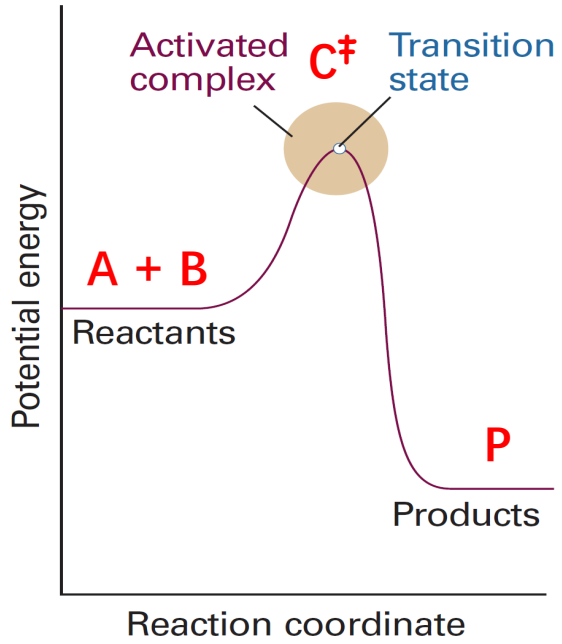
F is the freedom, the number of vibrational modes

The electronic contribution

$$q^{\mathrm{E}} = \sum_{\text{energy levels}} g_j \mathrm{e}^{-\beta\varepsilon_j}$$

Transition State Theory

$$A + B \rightarrow P \qquad v = k_2[A][B]$$



In a rapid pre-equilibrium

$$A + B \rightleftharpoons C^{\ddagger} \qquad K^{\ddagger} = \frac{p_{C^{\ddagger}} p^{\ominus}}{p_A p_B}$$

$$p_J = RT[J] \rightarrow [C^{\ddagger}] = \frac{RT}{p^{\ominus}} K^{\ddagger}[A][B]$$

$$C^{\ddagger} \rightarrow P \qquad v = k^{\ddagger}[C^{\ddagger}] \rightarrow \boxed{k_2 = \frac{RT}{p^{\ominus}} k^{\ddagger} K^{\ddagger}}$$

$$k^{\ddagger} = \kappa\nu \qquad K^{\ddagger} = \frac{N_A q^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT} \qquad \Delta E_0 = E_0(C^{\ddagger}) - E_0(A) - E_0(B)$$

$$K^{\ddagger} = \frac{kT}{h\nu} \bar{K}^{\ddagger} \qquad \bar{K}^{\ddagger} = \frac{N_A \bar{q}^{\ominus}_{C^{\ddagger}}}{q^{\ominus}_A q^{\ominus}_B} e^{-\Delta E_0/RT} \qquad \Delta^{\ddagger} G = -RT \ln \bar{K}^{\ddagger}$$

Eyring equation: $k_2 = \kappa \frac{kT}{h} \bar{K}^{\ddagger}_c$ **or** $k_2 = \kappa \frac{kT}{h} \frac{RT}{p^{\ominus}} e^{-\Delta^{\ddagger} G/RT}$